



A Second-Order Boundary Condition Capturing Method for Solving the Elliptic Interface Problems on Irregular Domains

Hyuntae Cho¹ · Heejae Han¹ · Byungjoon Lee² · Youngsoo Ha¹ · Myungjoo Kang¹

Received: 12 March 2019 / Revised: 1 July 2019 / Accepted: 19 July 2019
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Abstract

A second-order boundary condition capturing method is presented for the elliptic interface problem with jump conditions in the solution and its normal derivative. The proposed method is an extension of the work in Liu et al. (J Comput Phys 160(1):151–178, 2000) to a higher order. The motivation of proposed method is that the approximated value at the interface can be reconstructed by proper interpolation based on the level set representation from Gibou et al. (J Comput Phys 176(1):205–227, 2002). A second-order accurate method is constructed, both in the solution and its gradient, using second-order finite difference approximation. Several numerical results demonstrate that the proposed method is indeed second-order accurate in the solution and its gradient in the L^2 and L^∞ norms.

Keywords Elliptic interface problems · Irregular domains · Ghost fluid method · Level-set method · Finite difference method

✉ Myungjoo Kang
makng@snu.ac.kr

Hyuntae Cho
choant@snu.ac.kr

Heejae Han
hhj5575@snu.ac.kr

Byungjoon Lee
blee@catholic.ac.kr

Youngsoo Ha
youngamath@snu.ac.kr

¹ Department of Mathematical Sciences, Seoul National University, Seoul, Korea

² Department of Mathematics, The Catholic University of Korea, Bucheon, Korea

1 Introduction

We consider the following elliptic interface problem:

$$\begin{aligned}\nabla \cdot (\beta \nabla u) &= f \text{ on } \Omega \setminus \Gamma \\ [u] &= a \text{ on } \Gamma \\ [\beta u_n] &= b \text{ on } \Gamma \\ u &= g \text{ on } \partial\Omega,\end{aligned}\tag{1}$$

where Ω is a bounded domain, $\partial\Omega$ denotes its boundary, and Γ is the interface that divides Ω into disjoint subdomains Ω^+ and Ω^- . Here, β is a positive function that may have discontinuity across the interface, and $[q] = q^+ - q^-$ denotes the jump condition along the interface, with the superscripts “ $\pm$ ” referring to $\Omega^\pm$. In addition to the mathematical significance of the problem (1), the accurate solution and its gradient is crucial in several applications, such as electrostatics and fluid dynamics. The aim of this study is to introduce a numerical method based on the finite difference method for this problem.

The immersed interface method (IIM) [7], which was introduced by Leveque and Li, is a widely known method for solving (1). It provides second-order accurate numerical solutions in the L^∞ norm. The underlying principle is to use multivariable Taylor expansion to approximate the jump conditions with high-order accuracy. IIM has been applied to various fields [6,8,10,12] and was later extended in [9,20] with some modifications. However, the global second-order convergence of the gradient of the solution obtained by IIM is still unclear. For recent related work, see [1,11].

Extrapolation methods [2,13] are an alternative approach to solving (1). The boundary condition capturing method [13], which is also known as ghost fluid method (GFM), is a well-known example. In this method, fictitious points are introduced to impose sharp jump conditions. The resulting system is a symmetric positive-definite linear system that can be solved by efficient iterative methods such as multigrid, PCG. The limitation of this method is that the obtained numerical solution attains only first-order accuracy. Chern et al. [2] addressed this issue and introduced the coupling interface method (CIM), enhancing accuracy by quadratic interpolation. Later, Guittet et al. [5] proposed the Voronoi interface method, using the concept of GFM with Voronoi mesh.

Another method for solving (1) is the correction function method proposed by Marques et al. [14], where (1) is reformulated using correction functions. This enhances the accuracy of the numerical solution and its gradient up to fourth order in the L^2 norm. However, this method can be used only when β is constant. It has been extended to the case of piecewise constant β in [15]. Recently, Coco et al. [3] have developed a multigrid-based method for solving (1) that exhibits second-order convergence of the gradient in the L^1 and L^∞ norms.

In this study, a second-order extension of the work in Liu et al. [13] is presented. It is motivated by the fact that the approximated value of u at the interface Γ can be obtained by the method in [13]. The main contribution of this study is that the proposed method not only provides a second-order accuracy solution to (1) but also its gradient is second-order accurate. It should be noted that the second-order finite difference approximation in the proposed method is inspired by Gibou et al. [4].

The rest of the paper is organized as follows. In Sect. 2, the level set representation and GFM are briefly reviewed. Section 3 contains the details of the proposed method in 1D and 2D spaces, respectively. In Sect. 4, 2D numerical experiments containing complex domains are presented. The detailed 2D formulation for some cases is presented in Appendix.

2 Preliminaries

Herein, two methodologies on which the proposed method relies are summarized. First, the level set method is introduced, which is used to represent interfaces. Furthermore, it is indicated how GFM can be used to discretize the Poisson's equation under jump conditions.

2.1 Level Set Method

The level set method [16] is widely used to describe interfaces. It represents the interface as the zero-level set of a continuous function defined on a computational domain Ω . In addition to representing the interface, this method has the advantage of computing the geometric quantities of the interface and handling complex interface problems.

In this study, the level set representation is mainly used for describing an irregular interface Γ embedded in Ω , as in [13]. More precisely, a level set function ϕ is constructed whose zero-level set $\{x \in \mathbb{R}^n | \phi(x) = 0\}$ is the interface Γ ; moreover, $\Omega^+ = \{x \in \mathbb{R}^n | \phi(x) > 0\}$ and $\Omega^- = \{x \in \mathbb{R}^n | \phi(x) < 0\}$ denote the outside and inside of Γ , respectively. Using ϕ , it can be determined whether the interface lies between two adjacent grid points. For example, in 1D, $\phi(x_i)\phi(x_{i+1}) < 0$ implies that Γ intersects at some point x_I on the grid segment $[x_i, x_{i+1}]$, which can be recovered by a proper interpolation of x_i and x_{i+1} .

An important geometric quantity for the problem (1) is the normal to the interface. The normal vector $n = (n_x, n_y)$ can be computed by

$$n = (n_x, n_y) = \left(\frac{\phi_x}{\sqrt{\phi_x^2 + \phi_y^2}}, \frac{\phi_y}{\sqrt{\phi_x^2 + \phi_y^2}} \right). \quad (2)$$

Throughout the paper, ϕ_x and ϕ_y are approximated using the fourth-order central finite difference method.

2.2 Ghost Fluid Method

GFM by Liu et al. [13] is a first-order method resulting in a symmetric linear system. In [13], jump conditions are considered by adding a constant vector to the right-hand side of the linear system, thereby maintaining the symmetry of the coefficient matrix. Here, we will focus on how these jump conditions are used to approximate the value of u at the interface, which is not aligned with the grid.

GFM assumes that $[u] = a$, $[\beta u_n] = b$, and the normal vector n are defined at the grid points and are smoothly extended off the interface. In 1D, when there is no adjacent node across the interface, $(\beta u_x)_x$ in (1) is discretized by standard second-order finite differences:

$$\left[\beta_{i+\frac{1}{2}} \left(\frac{u_{i+1} - u_i}{\Delta x} \right) - \beta_{i-\frac{1}{2}} \left(\frac{u_i - u_{i-1}}{\Delta x} \right) \right] / \Delta x = f_i.$$

We now assume that there is an adjacent node across the interface. The discretization will be presented only when $x_i \in \Omega^-$ and $x_{i+1} \in \Omega^+$. In other cases, discretization is performed similarly. First, the location of the interface $x_I \in [x_i, x_{i+1}]$ is estimated using linear interpolation of the level set function ϕ . Let

$$x_I = x_i + \theta \Delta x \quad (3)$$

for

$$\theta = \frac{|\phi_i|}{|\phi_i| + |\phi_{i+1}|}. \quad (4)$$

The fundamental principle of GFM is to discretize the following jump conditions

$$\begin{aligned} [\beta u_x] &= (\beta^+ u_x^+ - \beta^- u_x^-)(x_I), \\ [u] &= u^+(x_I) - u^-(x_I) \end{aligned} \quad (5)$$

to obtain approximated values of u at x_I . Let u_I^+ and u_I^- denote the approximated values of u at x_I in Ω^+ and Ω^- , respectively. Similarly, β^+ and β^- are values of the function β at x_I in Ω^+ and Ω^- , respectively. The derivatives u_x^+ , u_x^- at x_I are approximated using the first order finite difference:

$$\begin{aligned} u_x^+ &\approx \frac{u_{i+1} - u_I^+}{(1 - \theta)\Delta x}, \\ u_x^- &\approx \frac{u_I^- - u_i}{\theta\Delta x}. \end{aligned}$$

Jumps at the interface are approximated by linear interpolation of the jumps at grid points:

$$\begin{aligned} [u] &\approx (1 - \theta)a_i + \theta a_{i+1} = a_I, \\ [\beta u_x] &\approx (1 - \theta)b_i + \theta b_{i+1} = b_I. \end{aligned}$$

Substituting these approximations into Eq. (5) yields

$$\begin{aligned} b_I &= \beta^+ \frac{u_{i+1} - u_I^+}{(1 - \theta)\Delta x} - \beta^- \frac{u_I^- - u_i}{\theta\Delta x}, \\ a_I &= u_I^+ - u_I^-. \end{aligned} \quad (6)$$

Solving for u_I^- , we obtain

$$u_I^- = \frac{1}{\hat{\beta}} (\beta^+ \theta u_{i+1} + \beta^- (1 - \theta) u_i - \beta^+ \theta a_I - b_I (1 - \theta) \theta \Delta x), \quad (7)$$

where $\hat{\beta} = \beta^+ \theta + \beta^- (1 - \theta)$. Using u_I^- , the 1D version of Eq. (1) is discretized as follows:

$$\left[\beta^- \frac{u_I^- - u_i}{\theta \Delta x} - \beta_{i-\frac{1}{2}} \frac{u_i - u_{i-1}}{\Delta x} \right] / \Delta x = f_i. \quad (8)$$

In 2D, to discretize jump conditions as in Eq. (6), the jump conditions $[\beta u_x]$, $[\beta u_y]$ should be approximated. For the normal vector $n = (n_x, n_y)$ and the tangent vector $\tau = (-n_y, n_x)$ defined at the interface, the following relation between $[\beta u_x]$, $[\beta u_y]$ and $[\beta u_n]$, $[\beta u_\tau]$ holds:

$$\begin{aligned} [\beta u_x] &= [\beta u_n] n_x - [\beta u_\tau] n_y, \\ [\beta u_y] &= [\beta u_n] n_y + [\beta u_\tau] n_x. \end{aligned} \quad (9)$$

In GFM, $[\beta u_x]$, $[\beta u_y]$ is approximated by neglecting $[\beta u_\tau]$:

$$\begin{aligned} [\beta u_x] &\approx [\beta u_n] n_x, \\ [\beta u_y] &\approx [\beta u_n] n_y. \end{aligned} \quad (10)$$

To discretize at a grid point (x_i, y_j) , we define four points $\mathbf{x}_R$, $\mathbf{x}_L$, $\mathbf{x}_T$, and $\mathbf{x}_B$ that either are grid points or lie on a grid segment. For example, $\mathbf{x}_R$ is defined as follows:

$$\mathbf{x}_R = (x_i + \theta_R \Delta x, y_j) \quad (11)$$

for

$$\theta_R = \begin{cases} 1 & \text{if } \phi_{i,j}\phi_{i+1,j} > 0 \\ \frac{|\phi_{i+1,j}|}{|\phi_{i,j}|+|\phi_{i+1,j}|} & \text{if } \phi_{i,j}\phi_{i+1,j} < 0 \end{cases}. \quad (12)$$

Thus, $\mathbf{x}_R = (x_{i+1}, y_j)$ if the grid segment does not intersect the interface, and $\mathbf{x}_R$ is on the grid segment if the interface lies between (x_i, y_j) and (x_{i+1}, y_j) . Likewise, the other three points are defined by

$$\begin{aligned} \mathbf{x}_L &= (x_i - \theta_L \Delta x, y_j), \\ \mathbf{x}_T &= (x_i, y_j + \theta_T \Delta y), \\ \mathbf{x}_B &= (x_i, y_j - \theta_B \Delta y) \end{aligned}$$

for similarly defined $\theta_L, \theta_T, \theta_B$ by linear interpolation of the level set function ϕ . Let u_R be the approximated value of u at $\mathbf{x}_R$. The expression of u_R differs depending on whether $\mathbf{x}_R$ is on the grid or not. Again, we assume that $(x_i, y_j) \in \Omega^-$.

When $\phi_{i,j}\phi_{i+1,j} > 0$ ($\mathbf{x}_R$ is a grid point),

$$\begin{aligned} \mathbf{x}_R &= (x_{i+1}, y_j), \\ u_R &= u_{i+1,j} \end{aligned} \quad (13)$$

and let

$$\beta_R^- = \beta_{i+\frac{1}{2},j}.$$

When $\phi_{i,j}\phi_{i+1,j} < 0$ ($\mathbf{x}_R$ is not a grid point), the jump condition is discretized as

$$\begin{aligned} [\beta u_x] &= (\beta_R^+ u_x^+ - \beta_R^- u_x^-)(\mathbf{x}_R), \\ [u] &= u^+(\mathbf{x}_R) - u^-(\mathbf{x}_R) \end{aligned} \quad (14)$$

to obtain an expression of u_R as in Eq. (7). Here, (10) is used to approximate $[\beta u_x]$ instead. β_R^+ and β_R^- denote the values of β at $\mathbf{x}_R$ in Ω^+ and Ω^- , respectively. A simple computation yields

$$u_R = \frac{1}{\hat{\beta}} (\beta_R^+ \theta_R u_{i+1,j} + \beta_R^- (1 - \theta_R) u_{i,j} - \beta_R^+ \theta_R a_R - b_R (1 - \theta_R) \theta_R \Delta x) \quad (15)$$

for

$$\hat{\beta} = \beta_R^+ \theta_R + \beta_R^- (1 - \theta_R)$$

and

$$\begin{aligned} a_R &= (1 - \theta_R) a_{i,j} + \theta_R a_{i+1,j}, \\ b_R &= (1 - \theta_R) b_{i,j}(n_x)_{i,j} + \theta_R b_{i+1,j}(n_x)_{i+1,j}. \end{aligned}$$

Similar expressions to (13) or (15) are derived for u_L, u_T, u_B and $\beta_L, \beta_T, \beta_B$. Subsequently, (1) is discretized as follows:

$$\begin{aligned} & \left[\beta_R^- \left(\frac{u_R - u_{i,j}}{\theta_R \Delta x} \right) - \beta_L^- \left(\frac{u_{i,j} - u_L}{\theta_L \Delta x} \right) \right] / \Delta x + \\ & \left[\beta_T^- \left(\frac{u_T - u_{i,j}}{\theta_T \Delta y} \right) - \beta_B^- \left(\frac{u_{i,j} - u_B}{\theta_B \Delta y} \right) \right] / \Delta y = f_{i,j}. \end{aligned}$$

3 Numerical Methods

In this section, a second-order GFM in 1D and 2D is presented. The main idea is to replace the first-order finite difference approximation of Eq. (6) by a second-order accurate formula. Among several candidates, the second-order finite difference approximation from [4] is selected, which originates in old work by Shortley and Weller [19].

This approximation is called the Shortley–Weller method, and the convergence properties of the solution and its gradient have been proved in several studies [18,21,22]. This section begins with an outline of the proposed method, which takes advantage of the Shortley–Weller method in 1D space.

3.1 One-Dimensional Case

We assume that the interface Γ lies between x_i, x_{i+1} . Let $x_I \in [x_i, x_{i+1}]$ be the approximated location of the interface defined by

$$x_I = x_i + \theta \Delta x \quad (16)$$

for

$$\theta_1 = \begin{cases} \frac{-D_{xx}^0 \phi + \sqrt{(D_{xx}^0 \phi)^2 - 4(D_{xx}^0 \phi) \phi_{i,j}}}{2D_{xx}^0 \phi} & \text{if } \phi_{i,j} < 0 \\ \frac{-D_{xx}^0 \phi - \sqrt{(D_{xx}^0 \phi)^2 - 4(D_{xx}^0 \phi) \phi_{i,j}}}{2D_{xx}^0 \phi} & \text{if } \phi_{i,j} > 0 \end{cases}, \quad (17)$$

where $D_{xx}^0 \phi = \frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{2}$ and $D_x^0 \phi = \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2}$. It should be noted that this approximation is formulated using quadratic interpolation of the level set function ϕ . Let u_I^+, u_I^- and β^+, β^- denote the approximated values of u and β at x_I in Ω^+ and Ω^- , respectively. Without loss of generality, we may assume that $\phi_i < 0$.

We approximate $[\beta u_x](x_I) \approx b_I$ by cubic interpolation of $b_{i-1}, b_i, b_{i+1}, b_{i+2}$, and $[u](x_I) \approx a_I$ by cubic interpolation of $a_{i-1}, a_i, a_{i+1}, a_{i+2}$. Subsequently, we use second-order finite differences to approximate u_x^- and u_x^+ at x_I :

$$\begin{aligned} u_x^- &\approx \left(\frac{\theta}{\theta+1} u_{i-1} - \frac{\theta+1}{\theta} u_i + \frac{2\theta+1}{\theta(\theta+1)} u_I^- \right) / \Delta x, \\ u_x^+ &\approx \left(-\frac{3-2\theta}{(1-\theta)(2-\theta)} u_I^+ + \frac{2-\theta}{1-\theta} u_{i+1} - \frac{1-\theta}{2-\theta} u_{i+2} \right) / \Delta x. \end{aligned} \quad (18)$$

Figure 1 shows the derivation of the second-order finite difference formula. Jump conditions are discretized by using the relation $[\beta u_x] = \beta^+ u_x^+ - \beta^- u_x^-$:

$$a_I = u_I^+ - u_I^-. \quad (19)$$

$$\begin{aligned} b_I &= \left(-\beta^+ \frac{3-2\theta}{(1-\theta)(2-\theta)} u_I^+ + \beta^+ \frac{2-\theta}{1-\theta} u_{i+1} - \beta^+ \frac{1-\theta}{2-\theta} u_{i+2} \right) / \Delta x \\ &\quad - \left(\beta^- \frac{\theta}{\theta+1} u_{i-1} - \beta^- \frac{\theta+1}{\theta} u_i + \beta^- \frac{2\theta+1}{\theta(\theta+1)} u_I^- \right) / \Delta x \end{aligned} \quad (20)$$

Substituting (19) into (20) yields the following compact representation for u_I^- :

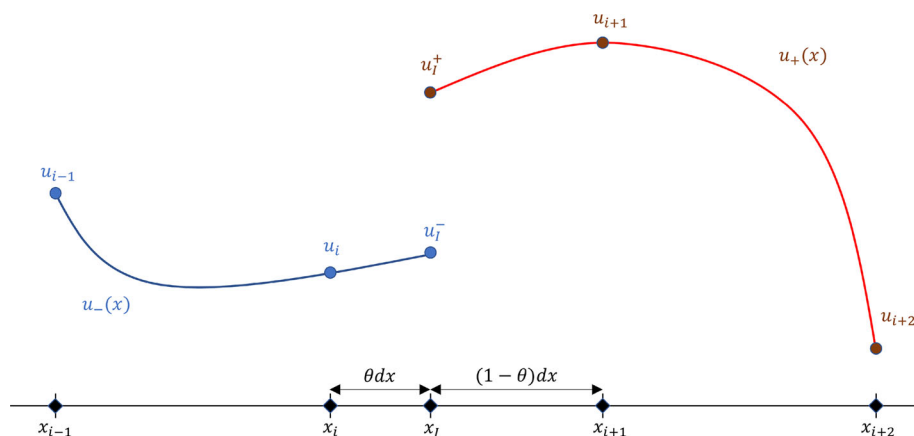


Fig. 1 Quadratic approximations for u_x^+ , u_x^- . Quadratic polynomials $u_-(x)$ and $u_+(x)$ are constructed interpolating u_{i-1} , u_i , u_I^- and u_I^+ , u_{i+1} , u_{i+2} , respectively. u_x^- and u_x^+ are approximated by u_I^- and u_I^+ , respectively

$$u_I^- = \left(-\beta^- \frac{\theta}{\theta+1} u_{i-1} + \beta^- \frac{\theta+1}{\theta} u_i + \beta^+ \frac{2-\theta}{1-\theta} u_{i+1} - \beta^+ \frac{1-\theta}{2-\theta} u_{i+2} \right) / \hat{\beta} - \left(\beta^+ \frac{3-2\theta}{(1-\theta)(2-\theta)} a_I + b_I \Delta x \right) / \hat{\beta}, \quad (21)$$

where

$$\hat{\beta} = \beta^+ \frac{3-2\theta}{(1-\theta)(2-\theta)} + \beta^- \frac{2\theta+1}{\theta(\theta+1)}.$$

Finally, (1) is discretized by the Shortley–Weller method using the u_I^- term in (21):

$$\left[\beta_{i+\frac{\theta}{2}} \left(\frac{u_I^- - u_i}{\theta \Delta x} \right) - \beta_{i-\frac{1}{2}} \left(\frac{u_i - u_{i-1}}{\Delta x} \right) \right] / \left(\frac{\theta+1}{2} \Delta x \right) = f_i.$$

A similar argument can be applied when Γ is located between x_i and x_{i-1} . In this case, x_I can be computed by $x_I = x_i - \theta \Delta x$, where θ is obtained by quadratic interpolation of the level set, as in (17). Let

$$\begin{aligned} \beta_1 &= \begin{cases} \beta^+(x_I) & \text{if } x_i \in \Omega^+ \\ \beta^-(x_I) & \text{if } x_i \in \Omega^- \end{cases}, \\ \beta_2 &= \begin{cases} \beta^-(x_I) & \text{if } x_i \in \Omega^+ \\ \beta^+(x_I) & \text{if } x_i \in \Omega^- \end{cases}, \\ \hat{\beta} &= \beta_2 \frac{3-2\theta}{(1-\theta)(2-\theta)} + \beta_1 \frac{2\theta+1}{\theta(\theta+1)}, \\ \mathbf{C} &= [\beta_1 \frac{\theta}{\theta+1} \quad -\beta_1 \frac{\theta+1}{\theta} \quad -\beta_2 \frac{2-\theta}{1-\theta} \quad \beta_2 \frac{1-\theta}{2-\theta}]. \end{aligned}$$

The 1D discretization may be summarized as follows:

Case 1 $\phi_i \phi_{i-1} > 0$ and $\phi_i \phi_{i+1} > 0$

$$\left[\beta_{i+\frac{1}{2}} \left(\frac{u_{i+1} - u_i}{\Delta x} \right) - \beta_{i-\frac{1}{2}} \left(\frac{u_i - u_{i-1}}{\Delta x} \right) \right] / \Delta x = f_i.$$

Case 2 $\phi_i \phi_{i+1} < 0$

$$u_I = -\frac{1}{\hat{\beta}} \mathbf{C} \begin{bmatrix} u_{i-1} \\ u_i \\ u_{i+1} \\ u_{i+2} \end{bmatrix} + \frac{\text{sgn}(\phi_i)}{\hat{\beta}} \left(\beta^+ \frac{3-2\theta}{(1-\theta)(2-\theta)} a_I + b_I \Delta x \right),$$

$$\left[\beta_{i+\frac{\theta}{2}} \left(\frac{u_I - u_i}{\theta \Delta x} \right) - \beta_{i-\frac{1}{2}} \left(\frac{u_i - u_{i-1}}{\Delta x} \right) \right] / \left(\frac{\theta+1}{2} \Delta x \right) = f_i.$$

Case 3 $\phi_i \phi_{i-1} < 0$

$$u_I = -\frac{1}{\hat{\beta}} \mathbf{C} \begin{bmatrix} u_{i+1} \\ u_i \\ u_{i-1} \\ u_{i-2} \end{bmatrix} + \frac{\text{sgn}(\phi_i)}{\hat{\beta}} \left(\beta^+ \frac{3-2\theta}{(1-\theta)(2-\theta)} a_I - b_I \Delta x \right),$$

$$\left[\beta_{i+\frac{1}{2}} \left(\frac{u_{i+1} - u_i}{\Delta x} \right) - \beta_{i-\frac{\theta}{2}} \left(\frac{u_i - u_{i-1}}{\theta \Delta x} \right) \right] / \left(\frac{\theta+1}{2} \Delta x \right) = f_i.$$

3.2 Two-Dimensional Case

The method can be extended to 2D in a dimension-by-dimension manner by approximating $[\beta u_x], [\beta u_y]$ in two different ways. One way is to use the relation between $[\beta u_x], [\beta u_y]$ and $[\beta u_n], [\beta u_\tau]$, the other is to use the definition of $[\beta u_x], [\beta u_y]$ combined with second-order finite differences in Cartesian directions. Then, the two approximations are equated to discretize the jump conditions.

As in Sect. 2.2, we first set four points near (x_i, y_j) , where the value of u will be approximated. Again, we assume that $(x_i, y_j) \in \Omega^-$. Let

$$\mathbf{x}_R = (x_R, y_j) = (x_i + \theta_R \Delta x, y_j) \quad (22)$$

for

$$\theta_R = \begin{cases} 1 & \text{if } \phi_{i,j} \phi_{i+1,j} > 0 \\ -\frac{D_{xx}^0 \phi + \sqrt{(D_{xx}^0 \phi)^2 - 4(D_{xx}^0 \phi) \phi_{i,j}}}{2D_{xx}^0 \phi} & \text{if } \phi_{i,j} \phi_{i+1,j} < 0 \end{cases}, \quad (23)$$

where $D_{xx}^0 \phi = \frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{2}$ and $D_x^0 \phi = \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2}$. When the interface intersects the grid segment connecting (x_i, y_j) and (x_{i+1}, y_j) , $\mathbf{x}_R$ approximates the intersection by quadratic interpolation of the level set function ϕ . Conversely, if (x_i, y_j) and (x_{i+1}, y_j) belong to same region, $\mathbf{x}_R = (x_{i+1}, y_j)$. Let

$$\begin{aligned} \mathbf{x}_L &= (x_L, y_j) = (x_i - \theta_L \Delta x, y_j), \\ \mathbf{x}_T &= (x_i, y_T) = (x_i, y_j + \theta_T \Delta y), \\ \mathbf{x}_B &= (x_i, y_B) = (x_i, y_j - \theta_B \Delta y) \end{aligned}$$

for similarly defined $\theta_L, \theta_T, \theta_B$. We also define $u_T^-, u_B^-, u_L^-, u_R^-$, which are the approximated values of u at $\mathbf{x}_T, \mathbf{x}_B, \mathbf{x}_L, \mathbf{x}_R$ in Ω^- , respectively. We assume that $[u] = a$, $[\beta u_n] = b$, and the normal vector $n = (n_x, n_y)$ and the tangent vector $\tau = (-n_y, n_x)$ are smoothly extended to the grid points on the computational domain.

3.2.1 Jump Conditions of Normal and Tangential Derivatives

We first present the approximation of $[\beta u_x]$, $[\beta u_y]$ by jump conditions of normal and tangential derivatives. Consider the exact relation (9):

$$\begin{pmatrix} [\beta u_x] \\ [\beta u_y] \end{pmatrix} = \begin{pmatrix} n_x & -n_y \\ n_y & n_x \end{pmatrix} \begin{pmatrix} [\beta u_n] \\ [\beta u_\tau] \end{pmatrix}.$$

For each Cartesian direction, we have

$$\begin{aligned} [\beta u_x] &= [\beta u_n]n_x - [\beta u_\tau]n_y \\ &= [\beta u_n]n_x - [\beta]u_\tau^- n_y - \beta^+[u]_\tau n_y \\ &= [\beta u_n]n_x - [\beta]u_\tau^+ n_y - \beta^-[u]_\tau n_y \end{aligned} \quad (24)$$

and

$$\begin{aligned} [\beta u_y] &= [\beta u_n]n_y + [\beta u_\tau]n_x \\ &= [\beta u_n]n_y + [\beta]u_\tau^- n_x + \beta^+[u]_\tau n_x \\ &= [\beta u_n]n_y + [\beta]u_\tau^+ n_x + \beta^-[u]_\tau n_x, \end{aligned} \quad (25)$$

where the following identity was used:

$$[pq] = [p]q^- + p^+[q].$$

To discretize $[\beta u_x]$, $[\beta u_y]$ from (24) and (25), it is essential to approximate $[u]_\tau$ and u_τ^- .

First, we define a_τ at each grid point by

$$(a_\tau)_{i,j} = -(D_x a)(n_y)_{i,j} + (D_y a)(n_x)_{i,j}$$

for $D_x a = \frac{-a_{i+2,j} + 8a_{i+1,j} - 8a_{i-1,j} + a_{i-2,j}}{12\Delta x}$ and $D_y a = \frac{-a_{i,j+2} + 8a_{i,j+1} - 8a_{i,j-1} + a_{i,j-2}}{12\Delta y}$. As $[u] = a$, a_τ can be seen as a natural approximation of $[u]_\tau$.

Next, let $\mathbf{x}_{\text{ext}} = (x_{\text{ext}}, y_{\text{ext}})$ be any of the four points (x_{i-1}, y_{j-1}) , (x_{i+1}, y_{j-1}) , (x_{i-1}, y_{j+1}) , (x_{i+1}, y_{j+1}) belonging to Ω^- , as it is unlikely that all points belong to Ω^+ . u_{ext} refers to the value of u at $\mathbf{x}_{\text{ext}}$. We consider the quadratic polynomial $P(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ that interpolates

$$\begin{aligned} P(\mathbf{x}_T) &= u_T^-, \\ P(\mathbf{x}_B) &= u_B^-, \\ P(\mathbf{x}_L) &= u_L^-, \\ P(\mathbf{x}_R) &= u_R^-, \\ P(x_i, y_j) &= u_{i,j}, \\ P(\mathbf{x}_{\text{ext}}) &= u_{\text{ext}}. \end{aligned} \quad (26)$$

Then the coefficients of P satisfy

$$\begin{bmatrix} (x_i)^2 & x_i y_T & (y_T)^2 & x_i & y_T & 1 \\ (x_i)^2 & x_i y_B & (y_B)^2 & x_i & y_B & 1 \\ (x_L)^2 & x_L y_j & (y_j)^2 & x_L & y_j & 1 \\ (x_R)^2 & x_R y_j & (y_j)^2 & x_R & y_j & 1 \\ (x_i)^2 & x_i y_j & (y_j)^2 & x_i & y_j & 1 \\ (x_{\text{ext}})^2 & x_{\text{ext}} y_{\text{ext}} & (y_{\text{ext}})^2 & x_{\text{ext}} & y_{\text{ext}} & 1 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} u_T^- \\ u_B^- \\ u_L^- \\ u_R^- \\ u_{i,j} \\ u_{\text{ext}} \end{bmatrix}. \quad (27)$$

The matrix on the left-hand side is always invertible. By solving the linear system (27), the coefficients of the polynomial P can be expressed as a linear combination of $u_T^-, u_B^-, u_L^-, u_R^-, u_{i,j}, u_{\text{ext}}$.

Finally, a_τ and $\nabla P \cdot \tau$ are used to approximate $[u]_\tau$ and u_τ^- in (24).

$$[\beta u_x] \approx b n_x - [\beta](\nabla P \cdot \tau) n_y - \beta^+ a_\tau n_y. \quad (28)$$

In the actual computation, $b, n_x, n_y, \beta, a_\tau$ are approximated by cubic interpolation of quantities defined at grid points.

3.2.2 Second Order Finite Differences in Cartesian Directions

Another approximation of $[\beta u_x]$ is obtained by applying the key idea of 3.1, which is to approximate u_x^- and u_x^+ . In 1D, when the interface lies in $[x_i, x_{i+1}]$, it can be ensured that no other interface point lies in $[x_{i-1}, x_{i+2}]$ if the grid is sufficiently refined. Thus, each approximation of u_x^- and u_x^+ in (18) is obtained by interpolation of u at x_{i-1}, x_i, x_I and x_I, x_{i+1}, x_{i+2} , respectively. In 2D, however, even in fine resolution, it cannot be ensured that the intersection of the interface and $[x_{i-1}, x_{i+2}] \times \{y_j\}$ is a single point. As shown in Fig. 2, such a case can occur even if the interface is smooth. Therefore, the discretization depends on the shape of the interface near (x_i, y_j) .

Case 1 $(x_{i+1}, y_j) \in \Omega^+, (x_{i+2}, y_j) \in \Omega^+$. Then the interface is located as in Fig. 3. This is the ideal case and we may use second-order finite differences in Cartesian directions to approximate u_x^- and u_x^+ at $\mathbf{x}_R$ as we did it in 1D:

$$\begin{aligned} u_x^+(\mathbf{x}_R) &\approx \left(-\frac{3-2\theta_R}{(1-\theta_R)(2-\theta_R)} u_R^+ + \frac{2-\theta_R}{1-\theta_R} u_{i+1,j} - \frac{1-\theta_R}{2-\theta_R} u_{i+2,j} \right) / \Delta x, \\ u_x^-(\mathbf{x}_R) &\approx \left(\frac{\theta_R}{\theta_L(\theta_L+\theta_R)} u_L^- - \frac{\theta_L+\theta_R}{\theta_L\theta_R} u_{i,j} + \frac{2\theta_R+\theta_L}{\theta_R(\theta_L+\theta_R)} u_R^- \right) / \Delta x. \end{aligned} \quad (29)$$

These approximations can be derived by replacing x_{i-1} with x_L in Fig. 1. Let a_R be an approximation of $[u](\mathbf{x}_R)$ computed by cubic interpolation of $a_{i-1,j}, a_{i,j}, a_{i+1,j}, a_{i+2,j}$. We approximate $\beta^+ u_x^+ - \beta^- u_x^-$ at $\mathbf{x}_R$ using (29) with the equation $u_R^+ - u_R^- = a_R$.

$$(\beta^+ u_x^+ - \beta^- u_x^-)|_{\mathbf{x}=\mathbf{x}_R} \approx \left(\mathbf{B}_R \begin{pmatrix} u_R^- \\ u_L^- \end{pmatrix} + \mathbf{C}_R \mathbf{u}_{\text{one-sided}} - \beta_R^+ \frac{3-2\theta_R}{(1-\theta_R)(2-\theta_R)} a_R \right) / \Delta x \quad (30)$$

for

$$\mathbf{B}_R = \begin{pmatrix} -\hat{\beta}_R & -\beta_R^- \frac{\theta_R}{(\theta_L+\theta_R)\theta_L} \end{pmatrix}$$

and

$$\mathbf{C}_R = \begin{pmatrix} \beta_R^- \frac{\theta_L+\theta_R}{\theta_L\theta_R} & \beta_R^+ \frac{2-\theta_R}{1-\theta_R} & -\beta_R^+ \frac{1-\theta_R}{2-\theta_R} \end{pmatrix},$$

where

$$\hat{\beta}_R = \beta_R^+ \frac{3-2\theta_R}{(1-\theta_R)(2-\theta_R)} + \beta_R^- \frac{2\theta_R+\theta_L}{\theta_R(\theta_R+\theta_L)}$$

and

$$\mathbf{u}_{\text{one-sided}} = (u_{i,j} \ u_{i+1,j} \ u_{i+2,j}).$$

Here, β_R^+ and β_R^- are the values of β at $\mathbf{x}_R$ in Ω^+ and Ω^- , respectively.

Case 2 $(x_{i+1}, y_j) \in \Omega^+$, $(x_{i+2}, y_j) \in \Omega^-$

We define a new point

$$\mathbf{x}_r = (x_r, y_j) = (x_{i+2} - \theta_r \Delta x, y_j) \quad (31)$$

for similarly defined θ_r by quadratic interpolation of the level set. We note that $\mathbf{x}_r$ is an approximation of an intersection of the interface and the line segment connecting (x_{i+1}, y_j) and (x_{i+2}, y_j) . Figure 4 shows the shape of the interface and the location of $\mathbf{x}_r$. Let u_r^- denote the approximated value of u at $\mathbf{x}_r$ in Ω^- .

The additional point $\mathbf{x}_r$ is considered because we cannot approximate u_x^+ at $\mathbf{x}_R$ without it. In case 2, $u_{i+2,j}$ was used to approximate u_x^+ . If $(x_{i+2}, y_j) \in \Omega^-$, $u_{i+2,j}$ cannot be used for approximation in Ω^+ . As a result of introducing $\mathbf{x}_r$, we should also consider jump conditions at $\mathbf{x}_r$.

We approximate u_x^+ and u_x^- at $\mathbf{x}_R$ and $\mathbf{x}_r$, respectively, using second-order finite differences in Cartesian directions:

$$\begin{aligned} u_x^-(\mathbf{x}_R) &\approx \left(\frac{\theta_R}{\theta_L(\theta_L + \theta_R)} u_L^- - \frac{\theta_L + \theta_R}{\theta_L \theta_R} u_{i,j} + \frac{2\theta_R + \theta_L}{\theta_R(\theta_L + \theta_R)} u_R^- \right) / \Delta x, \\ u_x^+(\mathbf{x}_R) &\approx - \left(\frac{3 - 2\theta_R - \theta_r}{(1 - \theta_R)(2 - \theta_R - \theta_r)} u_R^+ + \frac{2 - \theta_R - \theta_r}{(1 - \theta_R)(1 - \theta_r)} u_{i+1,j} \right. \\ &\quad \left. - \frac{1 - \theta_R}{(2 - \theta_R - \theta_r)(1 - \theta_r)} u_r^+ \right) / \Delta x, \\ u_x^-(\mathbf{x}_r) &\approx \left(-\frac{2\theta_r + 1}{\theta_r(\theta_r + 1)} u_R^- + \frac{\theta_r + 1}{\theta_r} u_{i+2,j} - \frac{\theta_r}{\theta_r + 1} u_{i+3,j} \right) / \Delta x, \\ u_x^+(\mathbf{x}_r) &\approx \left(\frac{1 - \theta_r}{(1 - \theta_R)(2 - \theta_R - \theta_r)} u_R^+ - \frac{2 - \theta_R - \theta_r}{(1 - \theta_R)(1 - \theta_r)} u_{i+1,j} \right. \\ &\quad \left. + \frac{3 - \theta_R - 2\theta_r}{(2 - \theta_R - \theta_r)(1 - \theta_r)} u_r^+ \right) / \Delta x. \end{aligned} \quad (32)$$

Figure 5 shows the derivation of these approximations. Let a_R and a_r be the approximated values of $[u]$ at $\mathbf{x}_R$ and $\mathbf{x}_r$, respectively. We substitute two discretizations $a_R = u_R^+ - u_R^-$ and $a_r = u_r^+ - u_r^-$ to approximate $\beta^+ u_x^+ - \beta^- u_x^-$.

$$\begin{aligned} \left(\begin{array}{c} (\beta^+ u_x^+ - \beta^- u_x^-)|_{\mathbf{x}=\mathbf{x}_R} \\ (\beta^+ u_x^+ - \beta^- u_x^-)|_{\mathbf{x}=\mathbf{x}_r} \end{array} \right) &\approx \left(\mathbf{B}_R \begin{pmatrix} u_L^- \\ u_R^- \end{pmatrix} + \mathbf{C}_R \mathbf{u}_{\text{one-sided}} \right. \\ &\quad \left. + \left(\begin{array}{c} -\beta_R^+ \frac{3-2\theta_R-\theta_r}{(1-\theta_R)(2-\theta_R-\theta_r)} a_R \\ \beta_r^+ \frac{3-\theta_R-2\theta_r}{(2-\theta_R-\theta_r)(1-\theta_r)} a_r \end{array} \right) \right) / \Delta x \end{aligned} \quad (33)$$

for

$$\mathbf{B}_R = \begin{pmatrix} -\beta_R^- \frac{\theta_R}{\theta_L(\theta_L + \theta_R)} & -\hat{\beta}_R & -\beta_R^+ \frac{1-\theta_R}{(2-\theta_R-\theta_r)(1-\theta_r)} \\ 0 & \beta_r^+ \frac{1-\theta_r}{(1-\theta_R)(2-\theta_R-\theta_r)} & \beta_r^- \end{pmatrix}$$

and

$$\mathbf{C}_R = \begin{pmatrix} \beta_R^- \frac{\theta_L + \theta_R}{\theta_L \theta_R} & \beta_R^+ \frac{2-\theta_R-\theta_r}{(1-\theta_R)(1-\theta_r)} & 0 & 0 \\ 0 & -\beta_r^+ \frac{2-\theta_R-\theta_r}{(1-\theta_R)(1-\theta_r)} & -\beta_r^- \frac{\theta_r + 1}{\theta_r} & \beta_r^- \frac{\theta_r}{\theta_r + 1} \end{pmatrix},$$

where

$$\mathbf{u}_{\text{one-sided}} = [u_i \ u_{i+1} \ u_{i+2} \ u_{i+3}]^T$$

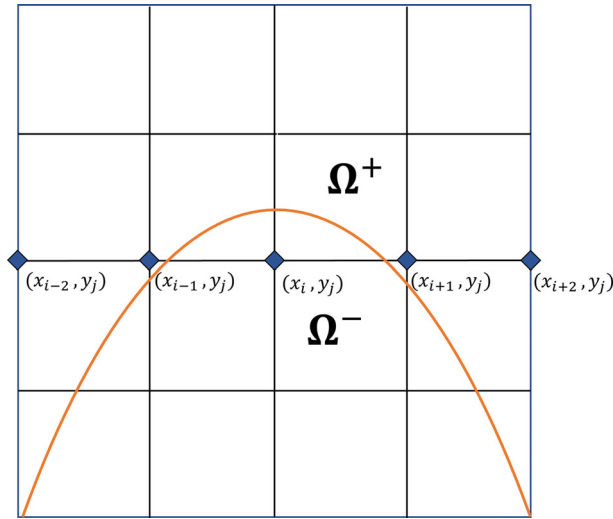


Fig. 2 Two-point intersection of the interface and $[x_{i-1}, x_{i+2}] \times \{y_j\}$

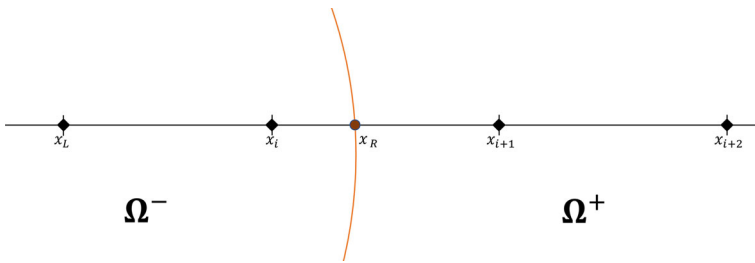


Fig. 3 Interface and grids for case 1

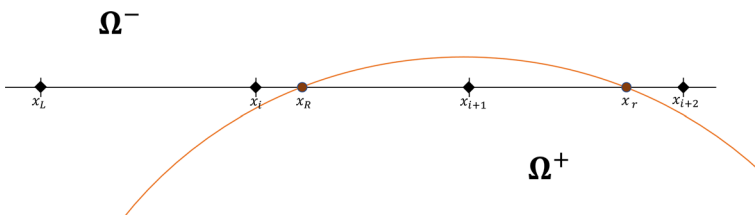


Fig. 4 Interface and grids for case 2

and

$$\hat{\beta}_R = \beta_R^- \frac{2\theta_R + \theta_L}{\theta_R(\theta_L + \theta_R)} + \beta_R^+ \frac{3 - 2\theta_R - \theta_r}{(1 - \theta_R)(2 - \theta_R - \theta_r)},$$

$$\hat{\beta}_r = \beta_r^- \frac{\theta_r + 1}{\theta_r(\theta_r + 1)} + \beta_r^+ \frac{3 - \theta_R - 2\theta_r}{(2 - \theta_r - \theta_R)(1 - \theta_r)}.$$

Again, $\beta_R^\pm$ and $\beta_r^\pm$ are the values of β at $\mathbf{x}_R$ and $\mathbf{x}_r$ in $\Omega^\pm$, respectively.

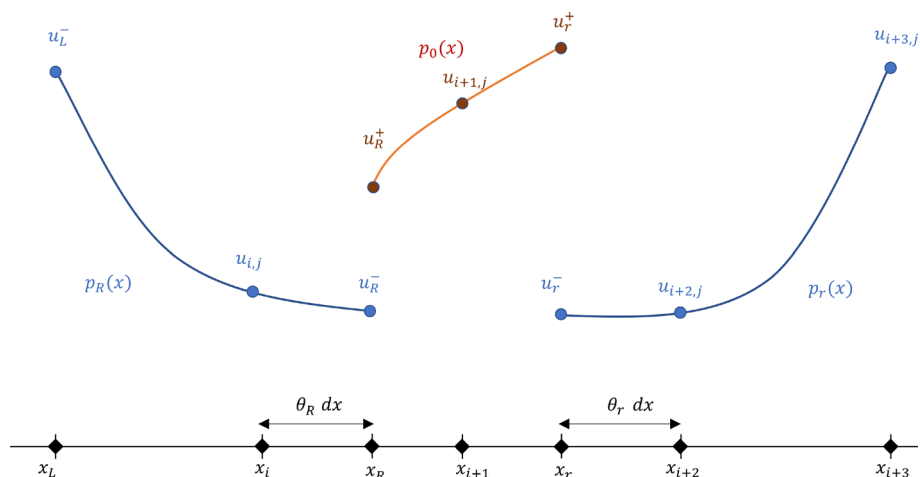


Fig. 5 Construction of quadratic polynomials $p_R(x)$ interpolating $u_L^-, u_{i,j}, u_R^-$, $p_0(x)$ interpolating $u_R^+, u_{i+1,j}, u_r^+$, and $p_r(x)$ interpolating $u_r^-, u_{i+2,j}, u_{i+3,j}$. u_x^- is approximated at $\mathbf{x}_R$ and $\mathbf{x}_r$ by $p'_R(x_R)$ and $p'_r(x_r)$, respectively. u_x^+ is approximated by p'_0

3.2.3 Discretizing Jump Conditions by Equating Two Different Approximations

We have presented two different approximation of $[\beta u_x]$ in Sects. 3.2.1 and 3.2.2. Jump condition will be discretized by equating two different approximations. As in Sect. 3.2.2, it depends on the shape of the interface, which is classified by sign of $\phi_{i+2,j}$.

Case I $(x_{i+1}, y_j) \in \Omega^+, (x_{i+2}, y_j) \in \Omega^+$

One approximation of $[\beta u_x]$ is formulated by approximating the quantities of (28) at $\mathbf{x}_R$. $[\beta u_x](\mathbf{x}_R)$ is denoted by $[\beta u_x]_R$.

$$[\beta u_x]_R \approx (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_R}. \quad (34)$$

Here, $\nabla P(\mathbf{x}_R) = (2Ax_R + By_j + D, Bx_R + 2Cy_j + E)$, and the other quantities are computed by cubic interpolation at $\mathbf{x}_R$ from the grid points (x_{i-1}, y_j) , (x_i, y_j) , (x_{i+1}, y_j) , (x_{i+2}, y_j) .

It should be noted that the constants A, B, C, D, E are linear combinations of $u_T^-, u_B^-, u_L^-, u_R^-, u_{\text{ext}}, u_{ij}$.

Another approximation of $[\beta u_x]_R$ is Eq. (30) which is derived by second order finite differences in Cartesian direction. We discretize jump condition by equating two approximations (30) and (34):

$$\begin{aligned} & (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_R} \\ &= \left(\mathbf{B}_R \begin{pmatrix} u_R^- \\ u_L^- \end{pmatrix} + \mathbf{C}_R \mathbf{u}_{\text{one-sided}} - \beta_R^+ \frac{3 - 2\theta_R}{(1 - \theta_R)(2 - \theta_R)} a_R \right) / \Delta x. \end{aligned} \quad (35)$$

Organizing the above discretization, we may rewrite it as follows:

$$\mathbf{M}_R \begin{pmatrix} u_R^- \\ u_L^- \\ u_T^- \\ u_B^- \end{pmatrix} = \mathbf{N}_R \mathbf{u}_R + d_R \quad (36)$$

for some $\mathbf{M}_R \in \mathbb{R}^{1 \times 4}$, $d_R \in \mathbb{R}$, $\mathbf{N}_R \in \mathbb{R}^{1 \times 4}$ and $\mathbf{u}_R = (u_{ij}, u_{\text{ext}}, u_{i+1,j}, u_{i+2,j})^T$.

Case 2 $(x_{i+1}, y_j) \in \Omega^+$, $(x_{i+2}, y_j) \in \Omega^-$

As in case 1, we approximate $[\beta u_x]$ at $\mathbf{x}_R$ and $\mathbf{x}_r$ by (28):

$$\begin{aligned} [\beta u_x]_R &\approx (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_R}, \\ [\beta u_x]_r &\approx (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_r}. \end{aligned} \quad (37)$$

Again, $\nabla P(\mathbf{x}_R) = (2Ax_R + By_j + D, Bx_R + 2Cy_j + E)$ and $\nabla P(\mathbf{x}_r) = (2Ax_r + By_j + D, Bx_r + 2Cy_j + E)$. The quantities at $\mathbf{x}_R$ were computed by cubic interpolation from (x_{i-1}, y_j) , (x_i, y_j) , (x_{i+1}, y_j) , (x_{i+2}, y_j) , and those at $\mathbf{x}_r$, by cubic interpolation from (x_i, y_j) , (x_{i+1}, y_j) , (x_{i+2}, y_j) , (x_{i+3}, y_j) .

The jump condition is discretized by equating (33) and (37):

$$\begin{aligned} &\left(\begin{aligned} (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_R} \\ (bn_x - [\beta](\nabla P \cdot \tau)n_y - \beta^+ a_\tau n_y)|_{\mathbf{x}=\mathbf{x}_r} \end{aligned} \right) \\ &= \mathbf{B}_R \begin{pmatrix} u_L^- \\ u_R^- \\ u_r^- \end{pmatrix} + \mathbf{C}_R \mathbf{u}_{\text{one-sided}} + \begin{pmatrix} -\beta_R^+ \frac{3-2\theta_R-\theta_r}{(1-\theta_R)(2-\theta_R-\theta_r)} a_R \\ \beta_r^+ \frac{3-\theta_R-2\theta_r}{(2-\theta_R-\theta_r)(1-\theta_r)} a_r \end{pmatrix}. \end{aligned} \quad (38)$$

Rearranging the equation yields

$$\mathbf{M}_R \begin{pmatrix} u_R^- \\ u_r^- \\ u_L^- \\ u_T^- \\ u_B^- \end{pmatrix} = \mathbf{N}_R \mathbf{u}_R + \mathbf{d}_R \quad (39)$$

for some $\mathbf{M}_R \in \mathbb{R}^{2 \times 5}$, $\mathbf{d}_R \in \mathbb{R}^{2 \times 1}$, $\mathbf{N}_R \in \mathbb{R}^{1 \times 5}$ and $\mathbf{u}_R = (u_{\text{ext}}, u_{ij}, u_{i+1,j}, u_{i+2,j}, u_{i+3,j})^T$.

If $(x_{i+1}, y_j) \in \Omega^-$, then $\mathbf{x}_R = (x_{i+1}, y_j)$. So jump conditions need not be considered. We simply discretize

$$u_R^- = u_{i+1,j}. \quad (40)$$

We note that for each $\mathbf{x}_R$, $\mathbf{x}_L$, $\mathbf{x}_T$ and $\mathbf{x}_B$, one of the Eqs. (36), (39), (40) must hold. Combining all the cases in general form, we obtain the linear system

$$\mathbf{M} \begin{pmatrix} u_R^- \\ u_r^- \\ u_L^- \\ u_l^- \\ u_T^- \\ u_t^- \\ u_B^- \\ u_b^- \end{pmatrix} = \mathbf{N}\mathbf{u} + \mathbf{D}, \quad (41)$$

where $\mathbf{u}$ is a vector whose components are functions u at grid points, and $\mathbf{D}$ is a constant. u_l^- , u_t^- , u_b^- and $\mathbf{x}_l$, $\mathbf{x}_t$, $\mathbf{x}_b$ are defined similarly to u_r^- and $\mathbf{x}_r$, respectively. It should be noted that the rows of (41) correspond to the equations at $\mathbf{x}_R$, $\mathbf{x}_r$, $\dots$, $\mathbf{x}_B$, $\mathbf{x}_b$. The coefficient matrix $\mathbf{M}$ is not an 8×8 full matrix because $\mathbf{x}_T$, $\mathbf{x}_B$, $\mathbf{x}_L$, $\mathbf{x}_R$ may be on the grid and $\mathbf{x}_t$, $\mathbf{x}_b$, $\mathbf{x}_l$, $\mathbf{x}_r$ may not be defined.

For example, if $\mathbf{x}_R$ is on the grid, then the equation at $\mathbf{x}_R$ is $u_R^- = u_{i+1,j}$. Then we may replace u_R^- by $u_{i+1,j}$ and move the first column of $\mathbf{M}$ to $\mathbf{N}$. If $\mathbf{x}_r$ is not defined, the row corresponding to the equation at $\mathbf{x}_r$ is $\mathbf{0}$, and therefore we may delete the row as well as the column corresponding to u_r^- . Usually, the size of the reduced $\mathbf{M}$ is at most 3×3 . In Appendix, we summarize the explicit formula for $\mathbf{M}$ in four possible cases. After solving system (41), we discretize (1) by

$$\begin{aligned} & \left[\beta_{i,j+\frac{\theta_T}{2}} \left(\frac{u_T^- - u_{i,j}}{\theta_T \Delta y} \right) - \beta_{i,j-\frac{\theta_B}{2}} \left(\frac{u_{i,j} - u_B^-}{\theta_B \Delta y} \right) \right] / \left(\frac{\theta_T + \theta_B}{2} \Delta y \right) \\ & + \left[\beta_{i+\frac{\theta_R}{2},j} \left(\frac{u_R^- - u_{i,j}}{\theta_R \Delta x} \right) - \beta_{i-\frac{\theta_L}{2},j} \left(\frac{u_{i,j} - u_L^-}{\theta_L \Delta x} \right) \right] / \left(\frac{\theta_R + \theta_L}{2} \Delta x \right) = f_{i,j}. \end{aligned}$$

Remark Assuming case 2, even though u_r^- is not required for discretizing $(\beta u_x)_x$, the linear system containing u_r^- should be solved. Moreover, the expression of u_R^- involves $u_{i+3,j}$. Thus, we introduce a simplified version of Eq. (39). If each domain Ω^+ , Ω^- is connected, we may approximate u_R^- by interpolation of u_L^- , $u_{i,j}$, $u_{i+2,j}$. We ignore $u_{i+1,j}$ and replace Eq. (39) by

$$u_R^- = \frac{\theta_R(\theta_R - 2)}{\theta_L(\theta_L + 2)} u_L^- - \frac{(\theta_R + \theta_L)(\theta_R - 2)}{2\theta_L} u_{i,j} + \frac{(\theta_R + \theta_L)\theta_R}{4 + 2\theta_L} u_{i+2,j}.$$

Thereby, the formulation of the linear system (41) is simpler, as the unknown value u_r^- is not involved. Furthermore, $u_{i+3,j}$ is not contained in the system. There was little difference between the numerical results of the simplified version and those of the original.

3.3 Comparison with CIM2

The basic approach of CIM2 and the proposed method is to discretize the jump condition of the derivatives in Cartesian directions. CIM2 uses u_{xx} at grid points to discretize the jump condition, whereas the proposed method uses the approximated u at the interface. In 1D,

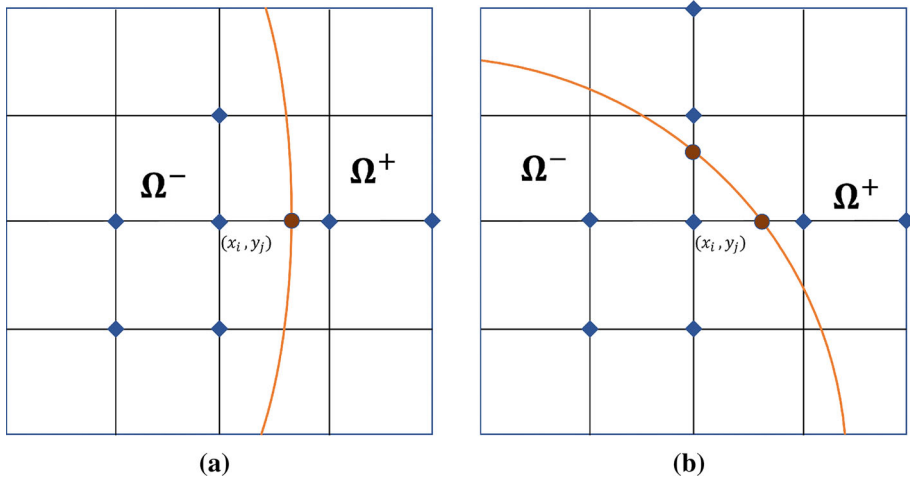


Fig. 6 Two examples of normal on-front grid point. Blue diamond-shaped points denote grid points, and red bullet points are intersections of the interface and grid segments (Color figure online)

replacing u_I^+ , u_I^- in Eq. (18) with $(u_{xx})_i$, $(u_{xx})_{i+1}$ by using the following discretizations yields the discretization of CIM2:

$$(u_{xx})_i = \left(\frac{u_I^- - u_i}{\theta \Delta x} - \frac{u_i - u_{i-1}}{\Delta x} \right) / \frac{\theta + 1}{2} \Delta x,$$

$$(u_{xx})_{i+1} = \left(\frac{u_{i+2} - u_{i+1}}{\Delta x} - \frac{u_i - u_I^+}{(1 - \theta) \Delta x} \right) / \frac{2 - \theta}{2} \Delta x.$$

In 2D, the authors in [2] note that CIM2 should be modified when the discretization contains non-normal on-front grid points, where a grid point $\mathbf{x}_{ij} = (x_i, y_j)$ near the interface Γ is said to be normal on-front if $\phi_{ij} \cdot \phi_{i\pm 1, j} < 0$ (or $\phi_{ij} \cdot \phi_{i, j\pm 1} < 0$) implies that $\phi_{ij} \cdot \phi_{i\mp 1, j} > 0$ (or $\phi_{ij} \cdot \phi_{i, j\mp 1} > 0$), as shown in Fig. 6. At normal on-front grid points, with proper choice of $\mathbf{x}_{\text{ext}}$, the proposed method yields the same discretization as CIM2.

Herein, we retain the fundamental idea of discretizing $[\beta u_x]$ and $[\beta u_y]$ using second-order finite differences on non-normal on-front grid points. Therefore, we claim that the proposed method is not only a second-order version of GFM but also an extension of CIM2 to arbitrary interfaces.

3.4 An Extension to Higher Dimension

An extension to higher dimension is straight forward. We will briefly explain how to extend the proposed method to 3 dimension. As in Sect. 3.2, define

$$\mathbf{X}_{i+} = (x_i + \theta_i \Delta x, y_j, z_k) \quad (42)$$

with quadratic interpolation of level-set ϕ . So $\mathbf{X}_{i+}$ is location of the interface if interface intersects grid segments connecting (x_i, y_j, z_k) and (x_{i+1}, y_j, z_k) , or is equal to (x_{i+1}, y_j, z_k) if (x_i, y_j, z_k) and (x_{i+1}, y_j, z_k) belongs to same region. $\mathbf{X}_{i-}$, $\mathbf{X}_{j+}$, $\mathbf{X}_{j-}$, $\mathbf{X}_{k+}$ and $\mathbf{X}_{k-}$ are defined in similar and natural way. Again, we assume $(x_i, y_j, z_k) \in \Omega^-$.

Suppose we have normal vector $n = (n_x, n_y, n_z)$ and two tangent vectors $\tau^1 = (\tau_x^1, \tau_y^1, \tau_z^1)$, $\tau^2 = (\tau_x^2, \tau_y^2, \tau_z^2)$ smoothly extended off the interface. Then we have the following relation that connects jump conditions of normal and tangential derivatives to jump conditions of Cartesian derivatives:

$$\begin{pmatrix} [\beta u_n] \\ [\beta u_{\tau^1}] \\ [\beta u_{\tau^2}] \end{pmatrix} = \begin{pmatrix} n_x & n_y & n_z \\ \tau_x^1 & \tau_y^1 & \tau_z^1 \\ \tau_x^2 & \tau_y^2 & \tau_z^2 \end{pmatrix} \begin{pmatrix} [\beta u_x] \\ [\beta u_y] \\ [\beta u_z] \end{pmatrix}.$$

Then $[\beta u_x]$ can be decomposed into jump conditions of normal and tangential derivatives as

$$\begin{aligned} [\beta u_x] &= [\beta u_n]n_x + [\beta u_{\tau^1}]\tau_x^1 + [\beta u_{\tau^2}]\tau_x^2 \\ &= [\beta u_n]n_x + ([\beta]u_{\tau^1}^- + \beta^+[u]_{\tau^1})\tau_x^1 + ([\beta]u_{\tau^2}^- + \beta^+[u]_{\tau^2})\tau_x^2. \end{aligned} \quad (43)$$

First, let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ be standard orthonormal basis in 3D. We pick l such that $|n \cdot e_l|$ has the minimum value among $l = 1, 2, 3$. For each normal n defined at grid points near interface, define $\tau^1 = n \times e_l$ and $\tau^2 = n \times \tau^1$. With enough grid refinement, this defines locally smooth tangent vector. As in 2D, we can define

$$(a_{\tau^1})_{i,j,k} = (D_x a)(\tau_x^1)_{i,j,k} + (D_y a)(\tau_y^1)_{i,j,k} + (D_z a)(\tau_z^1)_{i,j,k}$$

for $D_x a = \frac{-a_{i+2,j,k} + 8a_{i+1,j,k} - 8a_{i-1,j,k} + a_{i-2,j,k}}{12\Delta x}$, $D_y a = \frac{-a_{i,j+2,k} + 8a_{i,j+1,k} - 8a_{i,j-1,k} + a_{i,j-2,k}}{12\Delta y}$ and $D_z a = \frac{-a_{i,j,k+2} + 8a_{i,j,k+1} - 8a_{i,j,k-1} + a_{i,j,k-2}}{12\Delta z}$. a_{τ^2} is defined in similar way. Then a_{τ^1} , a_{τ^2} can be used to approximate $[u]_{\tau^1}$, $[u]_{\tau^2}$. Next, choose three more grid points that belongs to same region with $\mathbf{X}_{ijk} = (x_i, y_j, z_k)$. Let $(x_{i_1}, y_{j_1}, z_{k_1})$, $(x_{i_2}, y_{j_2}, z_{k_2})$, $(x_{i_3}, y_{j_3}, z_{k_3})$ denote additional three points. Under the condition that

$$\det \begin{pmatrix} i_1 - i & j_1 - j & k_1 - k \\ i_2 - i & j_2 - j & k_2 - k \\ i_3 - i & j_3 - j & k_3 - k \end{pmatrix} \neq 0,$$

$\mathbf{X}_{i^\pm}$, $\mathbf{X}_{j^\pm}$, $\mathbf{X}_{k^\pm}$, $\mathbf{X}_{ijk}$ with above three extra points defines effective quadratic interpolation in 3D. Especially in our numerical experiments, we have chosen i_l , j_l , k_l such that $|i_l - i|$, $|j_l - j|$, $|k_l - k| \leq 1$. $\nabla P \cdot \tau^l$ for $l = 1, 2$ can be used to approximate $u_{\tau^l}^-$ where P denotes the quadratic interpolant. Together with a_{τ^l} and $\nabla P \cdot \tau^l$, one can approximate $[\beta u_x]$ by (43).

Approximation of jump conditions by second-order finite differences in Cartesian direction is not different from Sect. 3.2.2. After equating (43) with second order finite differences in Cartesian directions, we obtain the following system.

$$\mathbf{M} \begin{pmatrix} u_{i^+} \\ u_{i^-} \\ u_{j^+} \\ u_{j^-} \\ u_{k^+} \\ u_{k^-} \end{pmatrix} = \mathbf{N} \mathbf{u} + \mathbf{D}. \quad (44)$$

After solving (44), discretize using Shortley–Weller method:

$$\begin{aligned} & \left[\beta_{i,j,k+\frac{\theta_k^+}{2}} \left(\frac{u_{k^+} - u_{i,j,k}}{\theta_{k^+} \Delta z} \right) - \beta_{i,j,k-\frac{\theta_k^-}{2}} \left(\frac{u_{i,j,k} - u_{k^-}}{\theta_{k^-} \Delta z} \right) \right] / \left(\frac{\theta_{k^+} + \theta_{k^-}}{2} \Delta z \right) + \\ & \left[\beta_{i,j+\frac{\theta_j^+}{2},k} \left(\frac{u_{j^+} - u_{i,j,k}}{\theta_{j^+} \Delta y} \right) - \beta_{i,j-\frac{\theta_j^-}{2},k} \left(\frac{u_{i,j,k} - u_{j^-}}{\theta_{j^-} \Delta y} \right) \right] / \left(\frac{\theta_{j^+} + \theta_{j^-}}{2} \Delta y \right) + \\ & \left[\beta_{i+\frac{\theta_i^+}{2},j,k} \left(\frac{u_{i^+} - u_{i,j,k}}{\theta_{i^+} \Delta x} \right) - \beta_{i-\frac{\theta_i^-}{2},j,k} \left(\frac{u_{i,j,k} - u_{i^-}}{\theta_{i^-} \Delta x} \right) \right] / \left(\frac{\theta_{i^+} + \theta_{i^-}}{2} \Delta x \right) = f_{i,j,k}. \end{aligned} \quad (45)$$

4 Numerical Experiments

In this section, we conduct numerical experiments to verify the second-order accuracy of the proposed method. As the resulting linear system is not symmetric, we chose the restarted GMRES with incomplete LU preconditioner [17] as a linear solver. Their L^2 and L^∞ errors in the solution and its gradient are presented in each example. When no adjacent node is across the interface, we use central differences to approximate the gradient:

$$\begin{aligned} u_x(x_i, y_j) &\approx \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x}, \\ u_y(x_i, y_j) &\approx \frac{u_{i,j+1} - u_{i,j-1}}{2\Delta y}. \end{aligned}$$

If there is an adjacent node across the interface, we solve (41), and then approximate:

$$\begin{aligned} u_x(x_i, y_j) &\approx \frac{-(\theta_R)^2 u_L + ((\theta_R)^2 - (\theta_L)^2) u_{i,j} + (\theta_L)^2 u_R}{\theta_L \theta_R (\theta_L + \theta_R) \Delta x}, \\ u_y(x_i, y_j) &\approx \frac{-(\theta_T)^2 u_B + ((\theta_T)^2 - (\theta_B)^2) u_{i,j} + (\theta_B)^2 u_T}{\theta_B \theta_T (\theta_B + \theta_T) \Delta y}. \end{aligned}$$

All the computations were carried out in C++, and implemented on a personal computer with 8 GB memory and a 3.4 GHz quad-core CPU.

4.1 Laplace Equation: Simple Example with $\beta = 1$

We first consider the Laplace equation $\Delta u = 0$ on $[-1, 1]^2$ with jump discontinuity across the interface in the normal derivative. More precisely, the interface Γ is defined by the zero-level set of $\phi(x, y) = \sqrt{x^2 + y^2} - 0.5$, and the jump conditions are $[u] = 0$, $[u_n] = 2$. The exact solution is

$$u(x, y) = \begin{cases} 1 & \text{if } \phi(x, y) < 0 \\ 1 + \log(2\sqrt{x^2 + y^2}) & \text{if } \phi(x, y) \geq 0 \end{cases}.$$

Figure 7 shows the solution on a 64×64 grid, and Table 1 shows the results of the convergence tests, confirming that the proposed method is second-order accurate both in the solution and its gradient.

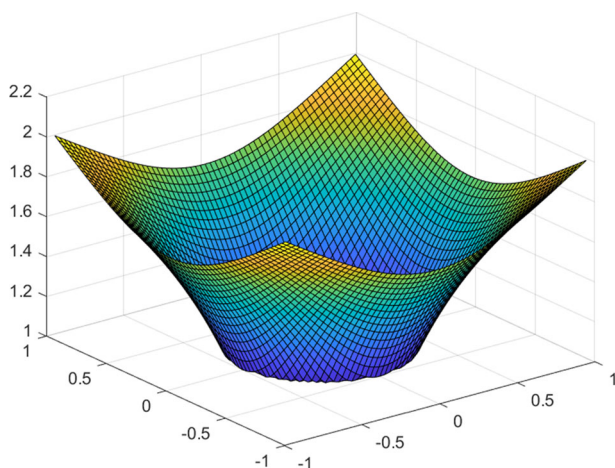


Fig. 7 Visualization of solution for Example 4.1

Table 1 Convergence of Example 4.1

Resolution	L^2 error in u	Order	L^2 error in ∇u	Order
32^2	2.35E-03		7.46E-03	
64^2	5.30E-04	2.15	1.78E-03	2.06
128^2	1.41E-04	1.91	4.71E-04	1.92
256^2	2.81E-05	2.33	1.07E-04	2.14
512^2	5.60E-06	2.33	2.48E-05	2.11
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order
32^2	2.23E-03		8.79E-03	
64^2	4.93E-04	2.18	2.51E-03	1.81
128^2	1.44E-04	1.78	8.18E-04	1.62
256^2	2.91E-05	2.31	2.61E-04	1.65
512^2	5.67E-06	2.36	5.94E-05	2.13

L^2 and L^∞ norms of solution and gradient are presented

4.2 Example with Irregular Interface Γ

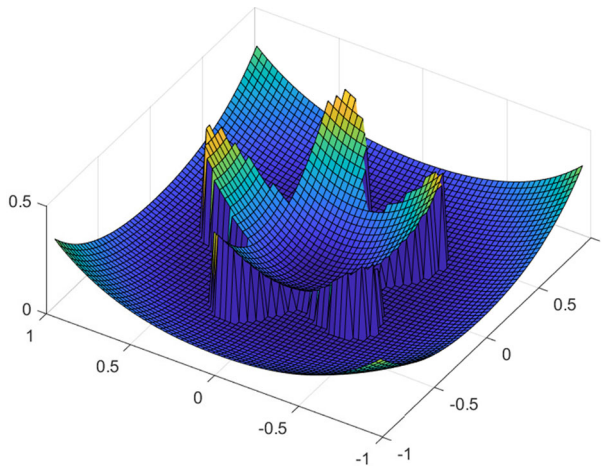
As a second example, we consider $\nabla \cdot (\beta \nabla u) = f$ on $\Omega = [-1, 1]^2$ with an irregular interface Γ embedded in the domain Ω . We choose the interface Γ to be the zero-level set of the function $\phi(x, y) = r - (0.5 + 0.15 \sin(5\theta))$ where $r = \sqrt{(x - 0.02\sqrt{5})^2 + (y - 0.02\sqrt{5})^2}$. Here, u is given by

$$u = \begin{cases} x^2 + y^2 & \text{if } \phi(x, y) < 0 \\ 0.1(x^2 + y^2)^2 - 0.01 \log(2\sqrt{x^2 + y^2}) & \text{if } \phi(x, y) \geq 0 \end{cases}$$

and f is given by

Table 2 Convergence of Example 4.2

Resolution	L^2 error in u	Order	L^2 error in ∇u	Order
32^2	1.84E-04		2.20E-03	
64^2	5.44E-05	1.76	5.19E-04	2.09
128^2	1.50E-05	1.86	1.23E-04	2.07
256^2	3.96E-06	1.93	3.06E-05	2.01
512^2	1.13E-06	1.81	7.40E-06	2.05
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order
32^2	3.31E-04		4.57E-03	
64^2	7.21E-05	2.20	1.46E-03	1.64
128^2	1.31E-05	2.46	3.16E-04	2.21
256^2	3.62E-06	1.86	4.94E-05	2.68
512^2	1.05E-06	1.79	1.09E-05	2.18

**Fig. 8** Visualization of solution for Example 4.2

$$f(x, y) = \begin{cases} 4 & \text{if } \phi(x, y) < 0 \\ 16(x^2 + y^2) & \text{if } \phi(x, y) \geq 0 \end{cases}$$

and β^+ , β^- are set to 10 and 1, respectively. The jump conditions are

$$\begin{aligned} [u] &= 0.1(x^2 + y^2)^2 - 0.01 \log(2\sqrt{x^2 + y^2}) - x^2 - y^2, \\ [\beta u_n] &= (4(x^2 + y^2) - 0.1(x^2 + y^2)^{-1} - 2)(xn_x + yn_y) \end{aligned}$$

where n_x and n_y are the x and y components, respectively, of the normal to the interface. As the exact values of n_x, n_y at the interface are not available, we approximate n_x, n_y using fourth-order accurate central finite differences using (2). The numerical accuracy results of the solution and its gradient are given in Table 2, and Fig. 8 shows the profile of the solution on a 64×64 grid. The results demonstrate that the accuracy of the proposed method is not significantly affected by interface complexity.

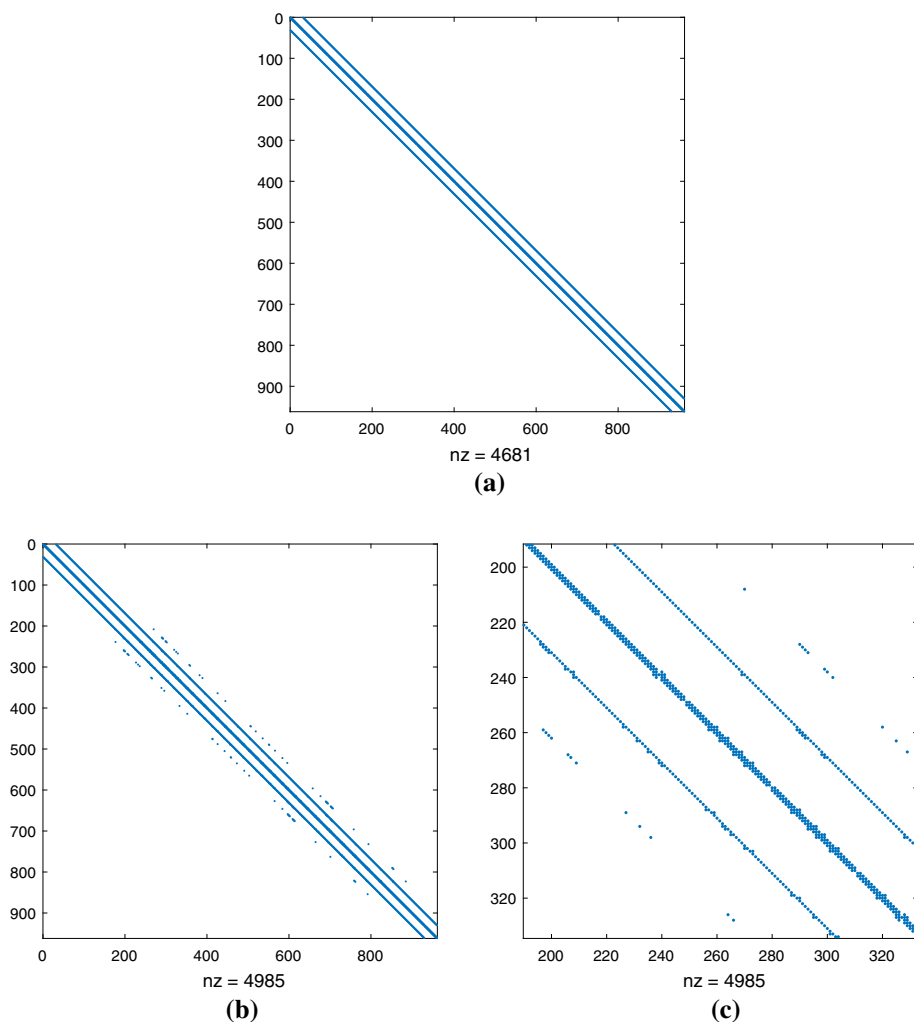


Fig. 9 Sparsity pattern of **a** the original GFM [13], **b** the proposed method and **c** the proposed method at the grid points near the interface (zoomed)

We also present a sparsity pattern of the proposed method and GFM [13] in Fig. 9. While each row of the matrix formed by [13] has at most 5 non-zero entry, the matrix obtained by the proposed method has 7, 8 or 9 non-zero entry at the rows corresponding to grid points near the interface. However, Table 3 shows that there is no significant difference between the growth rate of 1-norm condition number of the matrices obtained by two methods.

4.3 Example with General β

To verify that the accuracy of the proposed method is not heavily dependent on the coefficient β , we consider the problem (1) with a general function β . The other settings are the same as in Sect. 4.2 except for β and f , that is,

Table 3 1-Norm condition number of the matrices obtained by the proposed method and the original GFM [13] in Example 4.2

Resolution	The proposed method	The original GFM [13]
32^2	3.23E+03	1.52E+03
64^2	8.28E+03	6.01E+03
128^2	3.37E+04	2.40E+04
256^2	1.45E+05	9.57E+04
512^2	5.71E+05	3.83E+05

The condition number is higher in the proposed method, but the rate of growth as mesh is refined is similar to that of [13]

$$\beta(x, y) = \begin{cases} e^{10x} & \text{if } \phi(x, y) < 0 \\ \sin(x + y) + 2 & \text{if } \phi(x, y) \geq 0 \end{cases},$$

$$f(x, y) = \begin{cases} -2e^{10x} \sin(x + y) + 10e^{10x} \cos(x + y) & \text{if } \phi(x, y) < 0 \\ (\sin(x + y) + 2) \frac{4}{(x^2 + y^2 + 1)^2} + 2 \cos(x + y) \frac{x + y}{x^2 + y^2 + 1} & \text{if } \phi(x, y) \geq 0 \end{cases}.$$

The interface Γ is again the zero-level set of $\phi(x, y) = r - (0.5 + 0.15 \sin(5\theta))$. The exact solution for this problem is

$$u(x, y) = \begin{cases} \sin(x + y) & \text{if } \phi(x, y) < 0 \\ \log(x^2 + y^2 + 1) & \text{if } \phi(x, y) \geq 0 \end{cases}.$$

Jump conditions $[u], [\beta u_n]$ are computed according to given β and u . Table 4 shows the convergence rate for the solution and its gradient, and Fig. 10 shows the profile of the solution. It is noted that β was selected to have a large fluctuation in $[\beta]$ along the interface Γ . The results in Table 4 demonstrate that the proposed method is second-order accurate both in the solution and its gradient, regardless of the variance in $[\beta]$.

We remark that the accuracy of the gradient in the L^∞ norm is relatively low when the grid resolution is coarse. It appears that this is due to the large difference in β , which affects the computation of the gradient. In future work, we will present a mathematical analysis of the relationship between the jump discontinuity in β and the accuracy in ∇u .

4.4 Comparison with CIM2 in [2]

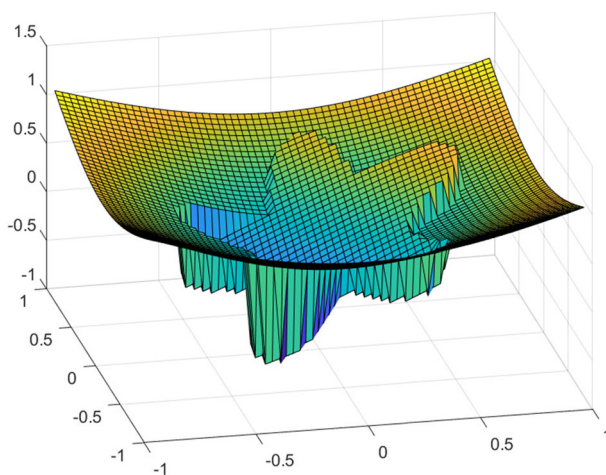
Herein, we compare the proposed method and CIM from [2]. We consider the problem (1) with $\beta^+ = 1$, $\beta^- = 2$ and the same jump conditions as in the previous example. Here, the function f is given by

$$f(x, y) = \begin{cases} -4 \sin(x + y) & \text{if } \phi(x, y) < 0 \\ 0 & \text{if } \phi(x, y) \geq 0 \end{cases}.$$

For comparison, we chose the interface Γ to be $\phi(x, y) = \sqrt{x^2 + y^2} - r$, where (a) $r = 0.5 + 0.0002$ (b) $r = 0.5 - \frac{2}{256} + 0.00001$. Figure 11 shows that cases (a) and (b) contain several non-normal on-front points on the 128×128 and 256×256 grids, respectively. In

Table 4 Convergence of Example 4.3

Resolution	L^2 error in u	Order	L^2 error in ∇u	Order
32^2	2.22E-01		6.52E-01	
64^2	5.93E-02	1.90	1.75E-01	1.90
128^2	1.29E-02	2.20	3.79E-02	2.20
256^2	3.60E-03	1.84	1.06E-02	1.84
512^2	7.46E-04	2.27	2.19E-03	2.27
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order
32^2	2.62E-01		1.55E+00	
64^2	6.87E-02	1.93	6.35E-01	1.29
128^2	1.47E-02	2.23	1.43E-01	2.15
256^2	4.11E-03	1.84	4.05E-02	1.82
512^2	8.41E-04	2.29	9.25E-03	2.13

**Fig. 10** Visualization of solution for Example 4.3

both cases, CIM2 should either make the resolution finer or take a hybrid CIM. It should be noted that the proposed method is not relevant to the previously mentioned points.

The accuracy results for both cases by the proposed method are given in Table 5. Here, N is the number of non-normal on-front points. This validates that the convergence of the proposed method is consistent, regardless of the interface location.

4.5 Example in 3D

We now consider example in 3D. Interface Γ is given as a sphere. In this example, we choose an interface Γ to be a sphere, which can be represented as a zero-levelset of

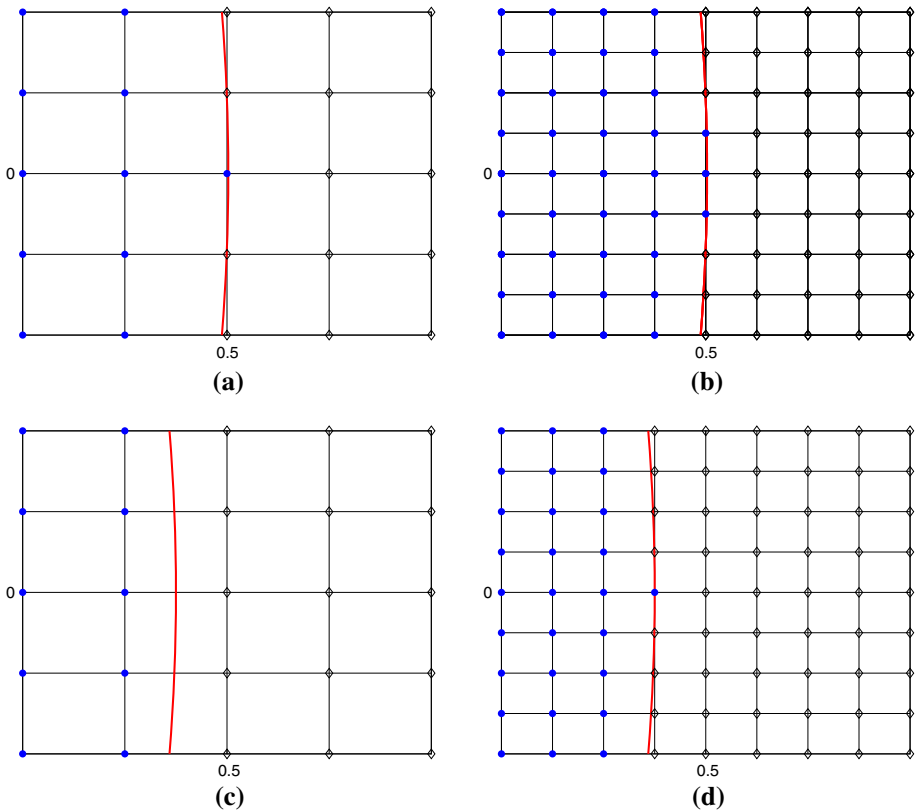


Fig. 11 Visualization of grid and level set of Example 4.4 near $(0.5, 0)$: **a** $r = 0.5 + 0.0002$ with resolution 128×128 , **b** $r = 0.5 + 0.0002$ with resolution 256×256 , **c** $r = 0.5 - \frac{2}{256} + 0.00001$ with resolution 128×128 and **d** $r = 0.5 - \frac{2}{256} + 0.00001$ with resolution 256×256 . Blue bullet points are grid points in Ω^- and black diamond-shaped points are grid points in Ω^+ (Color figure online)

$\phi(x, y, z) = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} - 0.5$. Similar to 4.1, solution u is taken from the Green's function of the Laplacian:

$$u = \begin{cases} 0.5 & \text{if } \phi(x, y, z) < 0 \\ \frac{1}{r} & \text{if } \phi(x, y, z) \geq 0 \end{cases}$$

where $r = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}$. $x_0 = 0.05\sqrt{5}$, $y_0 = 0.02\sqrt{3}$ and $z_0 = 0.01\sqrt{2}$ are chosen to avoid the symmetry of the interface around the origin. Cross section of the solution at $z = 0$ is displayed in Fig. 12. Note that $[u] = 0$ and $[\beta u_n] = \frac{\beta^+}{r^2}$. Here, we set β^+ and β^- by 1 and 100. Since solution is chosen as a Green's function of the Laplacian, $f = \beta \Delta u = 0$. Table 6 verifies the proposed method can be applied to 3D equation while maintaining the second-order convergence.

Table 5 Convergence of Example 4.4

(a)					
Resolution	L^2 error in u	Order	L^2 error in ∇u	Order	N
32^2	7.19E-03		1.98E-02		12
64^2	1.70E-03	2.08	4.75E-03	2.06	12
128^2	4.24E-04	2.00	1.19E-03	1.99	12
256^2	9.23E-05	2.20	2.74E-04	2.12	0
512^2	2.16E-05	2.10	6.63E-05	2.05	0
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order	N
32^2	6.54E-03		2.64E-02		12
64^2	1.62E-03	2.02	6.19E-03	2.09	12
128^2	4.10E-04	1.98	1.66E-03	1.90	12
256^2	9.28E-05	2.14	5.08E-04	1.71	0
512^2	2.21E-05	2.07	1.52E-04	1.74	0
(b)					
Resolution	L^2 error in u	Order	L^2 error in ∇u	Order	N
32^2	8.21E-03		2.17E-02		0
64^2	1.66E-03	2.31	4.74E-03	2.19	0
128^2	4.00E-04	2.05	1.18E-03	2.01	0
256^2	1.02E-04	1.97	2.92E-04	2.01	12
512^2	2.80E-05	1.87	7.83E-05	1.90	12
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order	N
32^2	7.55E-03		2.29E-02		0
64^2	1.63E-03	2.21	5.99E-03	1.93	0
128^2	4.05E-04	2.01	1.58E-03	1.92	0
256^2	1.00E-04	2.02	4.90E-04	1.69	12
512^2	2.77E-05	1.85	1.34E-04	1.87	12

N denotes the number of non-normal on-front grid points: (a) for $r = 0.5 + 0.0002$, and (b) for $r = 0.5 - \frac{2}{256} + 0.00001$

4.6 Example in 3D with Complex Interface

The last example is taken from [2]. We consider solution

$$u = \begin{cases} x^3 + xy^2 + y^3 + z^4 + \sin(3(x^2 + y^2)) & \text{if } \phi(x, y, z) < 0 \\ xy + x^4 + y^4 + xz^2 + \cos(2x + y^2 + z^3) & \text{if } \phi(x, y, z) \geq 0 \end{cases}$$

with peanut-shaped interface, which is given by zero-level set of level set function $\phi(r, \theta, \psi) = r - 0.5 - 0.2 \sin(2\theta) \sin(\psi)$ where (r, θ, ψ) is a spherical coordinate centered at $x_0 = 0.05\sqrt{5}$, $y_0 = 0.02\sqrt{3}$, $z_0 = 0.01\sqrt{2}$. Figure 13 depicts the interface and Fig. 14 shows the cross section of solution profile at $z = 0$. β^+ and β^- are given by 80 and 2

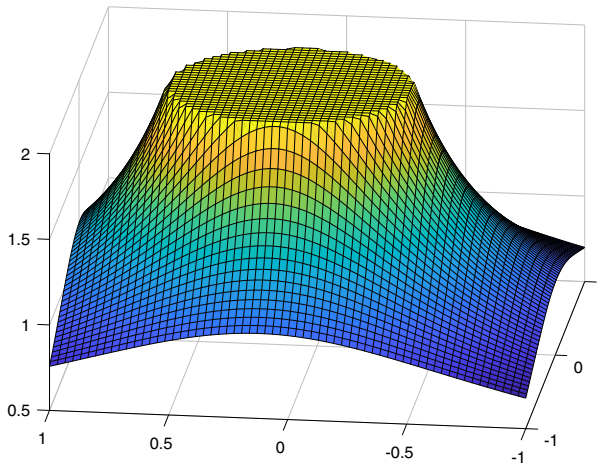


Fig. 12 Cross section of Example 4.5 at $z = 0$

Table 6 Convergence of Example 4.5

Resolution	L^2 error in u	Order	L^2 error in ∇u	Order
16^3	1.77E-01		3.08E-01	
32^3	3.96E-02	2.16	6.99E-02	2.14
64^3	1.00E-02	1.98	1.77E-02	1.98
128^3	2.50E-03	2.01	4.42E-03	2.00
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order
16^3	1.75E-01		5.09E-01	
32^3	3.77E-02	2.21	1.21E-01	2.07
64^3	9.29E-03	2.02	3.07E-02	1.99
128^3	2.30E-03	2.01	7.67E-03	2.00

respectively. Normal vector $n = (n_x, n_y, n_z)$ is set by (2) with 4th order central differences. $[u]$ and $[\beta u_n]$ is computed according to the solution. One can observe convergence of solution and its gradient in Table 7.

5 Conclusion

A second-order extension of GFM [13] was proposed. In 1D, the linear approximation of GFM was replaced by quadratic approximation, and the Shortley–Weller method was used to discretize Eq. (1). In 2D, two methods for approximating jump conditions of derivatives in Cartesian directions were introduced. One is to use the relation of jump conditions of the derivatives in normal and tangential directions, and the other is to use directly the definition of the jump condition. The problem (1) in 2D was discretized by equating the two approximations. Second-order convergence of both the solution and its gradient was observed in the L^2 and L^∞ norms with a complex interface. Furthermore, this methodology was extended to higher dimensions and the desired order of convergence was verified by numerical results.

Fig. 13 Visualization of the interface of Example 4.6

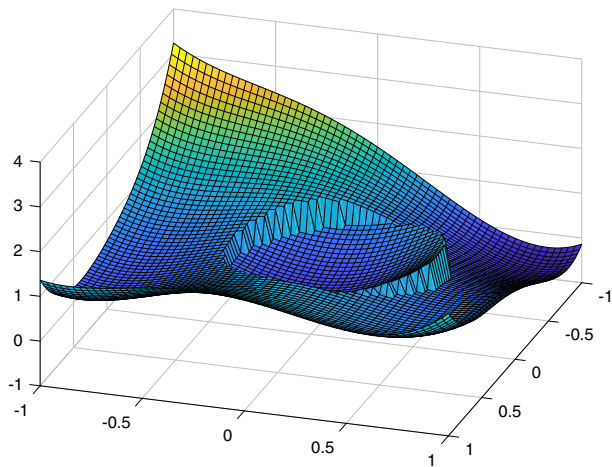
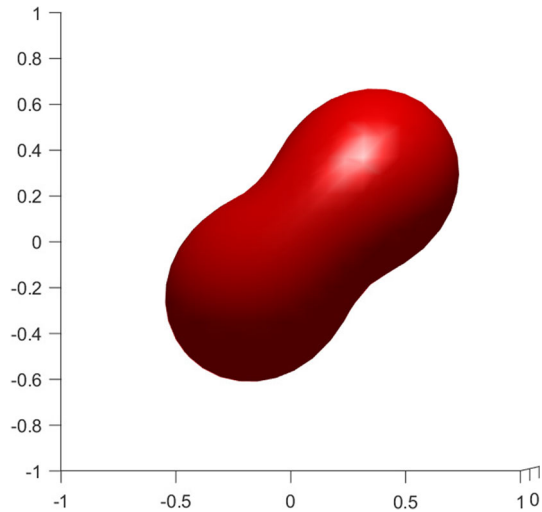


Fig. 14 Cross section of Example 4.6 at $z = 0$

Table 7 Convergence of Example 4.6

Resolution	L^2 error in u	Order	L^2 error in ∇u	Order
16^3	1.46E-02		9.41E-02	
32^3	3.52E-03	2.06	2.53E-02	1.90
64^3	8.78E-04	2.00	6.59E-03	1.94
128^3	2.19E-04	2.00	1.68E-03	1.97
Resolution	L^∞ error in u	Order	L^∞ error in ∇u	Order
16^3	1.69E-02		3.49E-01	
32^3	3.79E-03	2.16	8.19E-02	2.09
64^3	9.08E-04	2.06	2.88E-02	1.51
128^3	2.22E-04	2.03	6.73E-03	2.10

Although the convergence of the proposed method should be clarified and the restarted number of GMRES should be set higher when the ratio in β is high, it may be argued that the proposed method can be applied to physical phenomena where the accurate computation of the gradient is important.

Acknowledgements The research of Byungjoon Lee was supported by NRF Grant 2017R1C1B1008626 and POSCO Science Fellowship of POSCO TJ Park Foundation. The research of Myungjoo Kang was supported by the National Research Foundation of Korea (NRF)(2015R1A15A1009350, 2017R1A2A1A17069644).

Appendix: Formulation of $\mathbf{M}$ in (41)

The actual computation of the reduced $\mathbf{M}$ in (41) will herein be presented. We assume that $(x_i, y_j) \in \Omega^-$. With the choice of $\mathbf{x}_{\text{ext}}$, the approximation of ∇u at the interface by quadratic interpolation satisfying (26) will be formulated explicitly.

Case 1 The interface intersects one grid segment. Without loss of generality, we may assume that $\mathbf{x}_R$ is not on the grid, and $\mathbf{x}_T, \mathbf{x}_B, \mathbf{x}_L$ are on the grid, as shown in Fig. 15. We need only discretize the jump condition at $\mathbf{x}_R = (x_R, y_j)$. At the remaining points, $\mathbf{x}_T, \mathbf{x}_B$ and $\mathbf{x}_L$, we have $u_T^- = u_{i,j+1}$, $u_B^- = u_{i,j-1}$ and $u_L^- = u_{i-1,j}$. Let $\mathbf{x}_{\text{ext}} = (x_{i-1}, y_{j-1})$. Using second-order finite differences in Cartesian directions (29) with $u_L^- = u_{i-1,j}$, we obtain

$$u_x^+(\mathbf{x}_R) \approx \left(-\frac{3-2\theta_R}{(1-\theta_R)(2-\theta_R)}u_R^+ + \frac{2-\theta_R}{1-\theta_R}u_{i+1,j} - \frac{1-\theta_R}{2-\theta_R}u_{i+2,j} \right) / \Delta x,$$

$$u_x^-(\mathbf{x}_R) \approx \left(\frac{\theta_R}{\theta_R+1}u_{i-1,j} - \frac{\theta_R+1}{\theta_R}u_{i,j} + \frac{2\theta_R+1}{\theta_R(\theta_R+1)}u_R^- \right) / \Delta x.$$

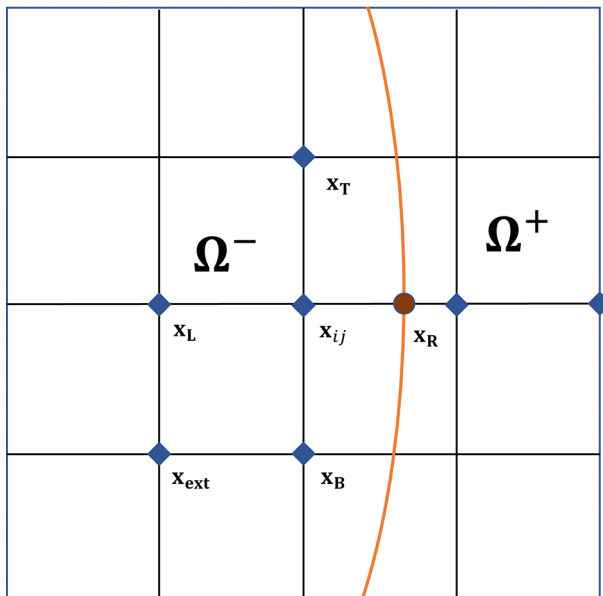


Fig. 15 Illustration for interface and points of case 1

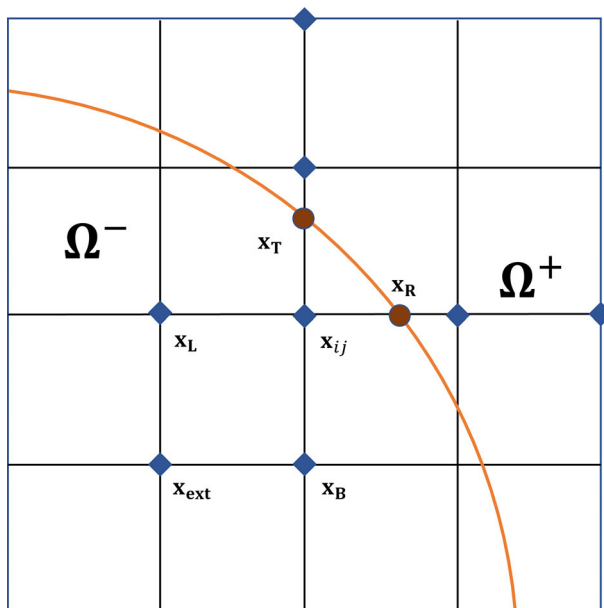


Fig. 16 Illustration for interface and points of case 2

We approximate $u_y^-(\mathbf{x}_R)$ by P_y , where P is obtained by quadratic interpolation (26).

$$u_y^-(\mathbf{x}_R) \approx \left(\theta_R(u_{i,j} - u_{i-1,j} - u_{i,j-1} + u_{i-1,j-1}) + \frac{1}{2}(u_{i,j+1} - u_{i,j-1}) \right) / \Delta y.$$

$[\beta u_x]_R$ is discretized by equating the following relations:

$$[\beta u_n]n_x - [\beta](-u_x^- n_y + u_y^- n_x)n_y - \beta^+[u]_\tau n_y = \beta^+ u_x^+ - \beta^- u_x^-.$$

$\mathbf{M}$ in (41) reduces to a 1×1 matrix, and we may write system (41) as follows:

$$\left(-\beta^+ \frac{3 - 2\theta_R}{(1 - \theta_R)(2 - \theta_r)} - (\beta^- + [\beta]n_y^2) \frac{2\theta_R + 1}{\theta_R(\theta_R + 1)} \right) u_R^- = \mathbf{N}\mathbf{u} + \mathbf{d}.$$

Case 2 The interface intersects two grid segment. We assume that $\phi_{i,j}\phi_{i+1,j} < 0$ and $\phi_{i,j}\phi_{i,j+1} < 0$. Thus, $\mathbf{x}_R$ and $\mathbf{x}_T$ are not on the grid, but $\mathbf{x}_B$ and $\mathbf{x}_L$ are on the grid. We choose $\mathbf{x}_{\text{ext}}$ to be (x_{i-1}, y_{j-1}) . The location of the interface and the choice of $\mathbf{x}_{\text{ext}}$ are shown in Fig. 16. As in case 1, we use second-order finite differences (29) and quadratic interpolation satisfying (26), and we approximate u_x^+, u_x^-, u_y^- at $\mathbf{x}_R$ and u_y^+, u_y^-, u_x^- at $\mathbf{x}_T$:

$$\begin{aligned} u_x^+(\mathbf{x}_R) &\approx \left(-\frac{3 - 2\theta_R}{(1 - \theta_R)(2 - \theta_r)} u_R^+ + \frac{2 - \theta_R}{1 - \theta_R} u_{i+1,j} - \frac{1 - \theta_R}{2 - \theta_R} u_{i+2,j} \right) / \Delta x, \\ u_x^-(\mathbf{x}_R) &\approx \left(\frac{\theta_R}{\theta_R + 1} u_{i-1,j} - \frac{\theta_R + 1}{\theta_R} u_{i,j} + \frac{2\theta_R + 1}{\theta_R(\theta_R + 1)} u_R^- \right) / \Delta x, \\ u_y^-(\mathbf{x}_R) &\approx \left(\theta_R u_{xy}^0 + u_y^0 \right) / \Delta y \end{aligned}$$

and

$$\begin{aligned} u_y^+(\mathbf{x}_T) &\approx \left(-\frac{3-2\theta_T}{(1-\theta_T)(2-\theta_T)} u_T^+ + \frac{2-\theta_T}{1-\theta_T} u_{i,j+1} - \frac{1-\theta_T}{2-\theta_T} u_{i,j+2} \right) / \Delta y, \\ u_y^-(\mathbf{x}_T) &\approx \left(\frac{\theta_T}{\theta_T+1} u_{i,j-1} - \frac{\theta_T+1}{\theta_T} u_{i,j} + \frac{2\theta_T+1}{\theta_T(\theta_T+1)} u_T^- \right) / \Delta y, \\ u_x^-(\mathbf{x}_T) &\approx \left(\theta_T u_{xy}^0 + u_x^0 \right) / \Delta x \end{aligned}$$

for

$$\begin{aligned} u_{xy}^0 &= (u_{i,j} - u_{i-1,j} - u_{i,j-1} + u_{i-1,j-1}), \\ u_x^0 &= \left(\frac{1}{\theta_R(\theta_R+1)} u_R^- + \frac{\theta_R-1}{\theta_R} u_{i,j} - \frac{\theta_R}{\theta_R+1} u_{i-1,j} \right), \\ u_y^0 &= \left(\frac{1}{\theta_T(\theta_T+1)} u_T^- + \frac{\theta_T-1}{\theta_T} u_{i,j} - \frac{\theta_T}{\theta_T+1} u_{i,j-1} \right). \end{aligned}$$

We discretize the jump conditions

$$[\beta u_n] n_x - [\beta](-u_x^- n_y + u_y^- n_x) n_y - \beta^+ [u]_\tau n_y = \beta^+ u_x^+ - \beta^- u_x^-$$

at $\mathbf{x}_R$ and

$$[\beta u_n] n_y + [\beta](-u_x^- n_y + u_y^- n_x) n_x + \beta^+ [u]_\tau n_x = \beta^+ u_y^+ - \beta^- u_y^-$$

at $\mathbf{x}_T$. The system (41) becomes

$$\mathbf{M} \begin{pmatrix} u_R^- \\ u_T^- \end{pmatrix} = \mathbf{N} \mathbf{u} + \mathbf{d}$$

for

$$\mathbf{M} = \begin{pmatrix} -\hat{\beta}_R & \left(\frac{n_x n_y [\beta]}{(\theta_T+1)\theta_T} \right) \Big|_{\mathbf{x}=\mathbf{x}_R} \\ \left(\frac{n_x n_y [\beta]}{(\theta_R+1)\theta_R} \right) \Big|_{\mathbf{x}=\mathbf{x}_T} & -\hat{\beta}_T \end{pmatrix}$$

where

$$\begin{aligned} \hat{\beta}_R &= \left(\beta^+ \frac{3-2\theta_R}{(1-\theta_R)(2-\theta_R)} + (\beta^- + [\beta] n_y^2) \frac{2\theta_R+1}{\theta_R(\theta_R+1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ \hat{\beta}_T &= \left(\beta^+ \frac{3-2\theta_R}{(1-\theta_R)(2-\theta_r)} + (\beta^- + [\beta] n_x^2) \frac{2\theta_R+1}{\theta_R(\theta_R+1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_T}. \end{aligned}$$

Case 3 The interface intersects two grid segments, but case 2 in Sect. 3.2 occurs. We again assume that $\mathbf{x}_R$ and $\mathbf{x}_T$ are not on the grid, but $\mathbf{x}_B$ and $\mathbf{x}_L$ are on the grid; however, we now have $(x_{i+2}, y_j) \in \Omega^-$. We define $\mathbf{x}_r$ as in (31). Figure 17 shows the interface and the location of $\mathbf{x}_r$. Again, quadratic interpolation (26) and second-order finite differences

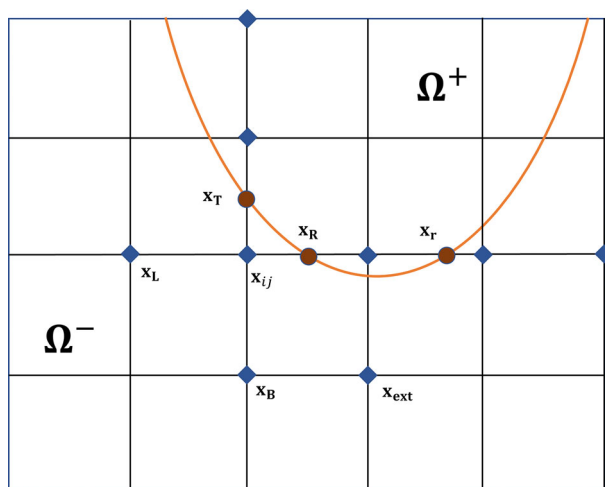


Fig. 17 Illustration for interface and points of case 3

in Cartesian direction lead to the approximations

$$\begin{aligned}
 u_x^+(\mathbf{x}_R) &\approx \left(-\frac{3-2\theta_R-\theta_r}{(1-\theta_R)(2-\theta_R-\theta_r)}u_R^+ + \frac{2-\theta_R-\theta_r}{(1-\theta_R)(1-\theta_r)}u_{i,j} \right. \\
 &\quad \left. - \frac{1-\theta_R}{(1-\theta_r)(2-\theta_R-\theta_r)}u_r^+ \right) / \Delta x, \\
 u_x^-(\mathbf{x}_R) &\approx \left(\frac{\theta_R}{\theta_R+1}u_{i-1,j} - \frac{\theta_R+1}{\theta_R}u_{i,j} + \frac{2\theta_R+1}{\theta_R(\theta_R+1)}u_R^- \right) / \Delta x, \\
 u_y^-(\mathbf{x}_R) &\approx \left(\theta_R u_{xy}^0 + u_y^0 \right) / \Delta y
 \end{aligned}$$

at $\mathbf{x}_R$,

$$\begin{aligned}
 u_x^+(\mathbf{x}_r) &\approx \left(\frac{1-\theta_r}{(1-\theta_R)(2-\theta_R-\theta_r)}u_R^+ - \frac{2-\theta_R-\theta_r}{(1-\theta_R)(1-\theta_r)}u_{i,j} \right. \\
 &\quad \left. + \frac{3-\theta_R-2\theta_r}{(1-\theta_r)(2-\theta_R-\theta_r)}u_r^+ \right) / \Delta x, \\
 u_x^-(\mathbf{x}_r) &\approx \left(-\frac{\theta_r}{\theta_r+1}u_{i+3,j} + \frac{\theta_r+1}{\theta_r}u_{i+2,j} - \frac{2\theta_r+1}{\theta_r(\theta_r+1)}u_r^- \right) / \Delta x, \\
 u_y^-(\mathbf{x}_r) &\approx \left((2-\theta_r)u_{xy}^0 + u_y^0 \right) / \Delta y
 \end{aligned}$$

at $\mathbf{x}_r$, and

$$\begin{aligned}
 u_y^+(\mathbf{x}_T) &\approx \left(-\frac{3-2\theta_T}{(1-\theta_T)(2-\theta_T)}u_T^+ + \frac{2-\theta_T}{1-\theta_T}u_{i,j+1} - \frac{1-\theta_T}{2-\theta_T}u_{i,j+2} \right) / \Delta y, \\
 u_y^-(\mathbf{x}_T) &\approx \left(\frac{\theta_T}{\theta_T+1}u_{i,j-1} - \frac{\theta_T+1}{\theta_T}u_{i,j} + \frac{2\theta_T+1}{\theta_T(\theta_T+1)}u_T^- \right) / \Delta y, \\
 u_x^-(\mathbf{x}_T) &\approx \left(\theta_T u_{xy}^0 + u_x^0 \right) / \Delta x
 \end{aligned}$$

at $\mathbf{x}_T$ for

$$\begin{aligned} u_{xy}^0 &= \left(\frac{2}{(\theta_R + 1)\theta_R} u_R + \frac{\theta_R - 2}{\theta_R} u_{i,j} - \frac{\theta_R - 1}{\theta_R + 1} u_{i-1,j} - u_{i+1,j-1} + u_{i,j-1} \right), \\ u_x^0 &= \left(\frac{1}{\theta_R(\theta_R + 1)} u_R^- + \frac{\theta_R - 1}{\theta_R} u_{i,j} - \frac{\theta_R}{\theta_R + 1} u_{i-1,j} \right), \\ u_y^0 &= \left(\frac{1}{\theta_T(\theta_T + 1)} u_T^- + \frac{\theta_T - 1}{\theta_T} u_{i,j} - \frac{\theta_T}{\theta_T + 1} u_{i,j-1} \right). \end{aligned}$$

By discretization of the jump conditions at $\mathbf{x}_T$, $\mathbf{x}_R$, and $\mathbf{x}_r$, we obtain the system

$$\mathbf{M} \begin{pmatrix} u_R^- \\ u_r^- \\ u_T^- \end{pmatrix} = \mathbf{N} \mathbf{u} + \mathbf{d}$$

for

$$\mathbf{M} = (m_{ij})_{i,j=1,2,3}$$

where

$$\begin{aligned} m_{11} &= \left(\beta^+ \frac{(1 - \theta_r)(3 - 2\theta_R - \theta_r)}{\hat{\theta}} + (\beta^- + [\beta]n_y^2) \frac{2\theta_R + 1}{\theta_R(\theta_R + 1)} \right. \\ &\quad \left. - [\beta]n_x n_y \frac{2\theta_R}{\theta_R(\theta_R + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{12} &= \left(\beta^+ \frac{(1 - \theta_R)^2}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{13} &= \left(-[\beta]n_x n_y \frac{1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{21} &= \left(\beta^+ \frac{(1 - \theta_r)^2}{\hat{\theta}} + [\beta]n_x n_y \frac{4 - 2\theta_r}{\theta_R(\theta_R + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_r}, \\ m_{22} &= \left(\beta^+ \frac{(3 - \theta_R - 2\theta_r)(1 - \theta_R)}{\hat{\theta}} + (\beta^- + [\beta]n_y^2) \frac{2\theta_r + 1}{\theta_r(\theta_r + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_r}, \\ m_{23} &= \left([\beta]n_x n_y \frac{1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_r}, \\ m_{31} &= \left(-[\beta]n_x n_y \frac{2\theta_T + 1}{\theta_R(\theta_R + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_T}, \\ m_{32} &= 0, \\ m_{33} &= \left(\beta^+ \frac{3 - 2\theta_T}{(1 - \theta_T)(2 - \theta_T)} + (\beta^- + [\beta]n_x^2) \frac{2\theta_T + 1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_T} \end{aligned}$$

and

$$\hat{\theta} = (1 - \theta_r)(2 - \theta_R - \theta_r)(1 - \theta_R).$$

Case 4 The interface intersects three grid segments.

Without loss of generality, we may assume that $\mathbf{x}_T$, $\mathbf{x}_L$, $\mathbf{x}_R$ are not on the grid, and $\mathbf{x}_B = (x_i, y_{j-1})$, as shown in Fig. 18. We choose $\mathbf{x}_{\text{ext}} = (x_{i-1}, y_{j-1})$. By quadratic interpolation (26) and second-order finite differences, at $\mathbf{x}_R$, we have

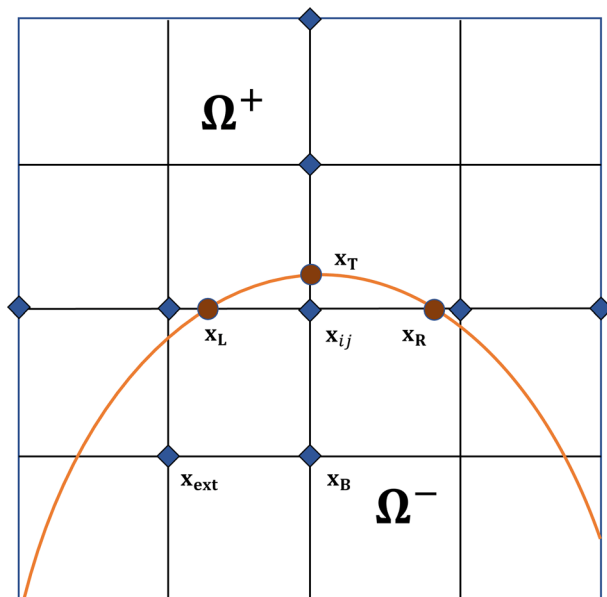


Fig. 18 Illustration for interface and points of case 4

$$\begin{aligned}
 u_x^+(\mathbf{x}_R) &\approx \left(-\frac{1-\theta_R}{2-\theta_R} u_{i+2,j}^+ + \frac{2-\theta_R}{1-\theta_R} u_{i+1,j} - \frac{3-2\theta_R}{(1-\theta_R)(1-\theta_R)} u_R^+ \right) / \Delta x, \\
 u_x^-(\mathbf{x}_R) &\approx \left(\frac{2\theta_R+\theta_L}{\theta_R(\theta_R+\theta_L)} u_R^- - \frac{\theta_R+\theta_L}{\theta_R\theta_L} u_{i,j} + \frac{\theta_R}{(\theta_R+\theta_L)\theta_L} u_L^- \right) / \Delta x, \\
 u_y^-(\mathbf{x}_R) &\approx \left(\theta_R u_{xy}^0 + u_y^0 \right) / \Delta y.
 \end{aligned}$$

At $\mathbf{x}_L$, we have

$$\begin{aligned}
 u_x^+(\mathbf{x}_L) &\approx \left(\frac{3-2\theta_L}{(1-\theta_L)(1-\theta_L)} u_L^+ - \frac{2-\theta_L}{1-\theta_L} u_{i-1,j} + \frac{1-\theta_L}{2-\theta_L} u_{i-2,j} \right) / \Delta x, \\
 u_x^-(\mathbf{x}_L) &\approx \left(-\frac{\theta_L}{\theta_R(\theta_R+\theta_L)} u_R^- + \frac{\theta_R+\theta_L}{\theta_R\theta_L} u_{i,j} - \frac{\theta_R+2\theta_L}{(\theta_R+\theta_L)\theta_L} u_L^- \right) / \Delta x, \\
 u_y^-(\mathbf{x}_L) &\approx \left(-\theta_L u_{xy}^0 + u_y^0 \right) / \Delta y
 \end{aligned}$$

and at $\mathbf{x}_T$, we have

$$\begin{aligned}
 u_y^+(\mathbf{x}_T) &\approx \left(-\frac{3-2\theta_T}{(1-\theta_T)(2-\theta_T)} u_T^+ + \frac{2-\theta_T}{1-\theta_T} u_{i,j+1} - \frac{1-\theta_T}{2-\theta_T} u_{i,j+2} \right) / \Delta y, \\
 u_y^-(\mathbf{x}_T) &\approx \left(\frac{\theta_T}{\theta_T+1} u_{i,j-1} - \frac{\theta_T+1}{\theta_T} u_{i,j} + \frac{2\theta_T+1}{\theta_T(\theta_T+1)} u_T^- \right) / \Delta y, \\
 u_x^-(\mathbf{x}_T) &\approx \left(\theta_T u_{xy}^0 + u_x^0 \right) / \Delta x
 \end{aligned}$$

for

$$\begin{aligned} u_{xy}^0 &= \left(\frac{\theta_L - 1}{(\theta_R + \theta_L)\theta_R} u_R^- + \frac{\theta_R - \theta_L + 1}{\theta_R \theta_L} u_{i,j} - \frac{\theta_R + 1}{(\theta_R + \theta_L)\theta_L} u_L - u_{i,j-1} + u_{i-1,j-1} \right), \\ u_x^0 &= \left(\frac{\theta_L}{\theta_R(\theta_R + \theta_L)} u_R^- + \frac{\theta_R - \theta_L}{\theta_R \theta_L} u_{i,j} - \frac{\theta_R}{(\theta_R + \theta_L)\theta_L} u_L^- \right), \\ u_y^0 &= \left(\frac{1}{\theta_T(\theta_T + 1)} u_T^- + \frac{\theta_T - 1}{\theta_T} u_{i,j} - \frac{\theta_T}{\theta_T + 1} u_{i,j-1} \right). \end{aligned}$$

We discretize the jump conditions

$$[\beta u_n]n_x - [\beta](-u_x^- n_y + u_y^- n_x)n_y - \beta^+[u]_\tau n_y = \beta^+ u_x^+ - \beta^- u_x^-$$

at $\mathbf{x}_L$, $\mathbf{x}_R$, and

$$n_y + [\beta](-u_x^- n_y + u_y^- n_x)n_x + \beta^+[u]_\tau n_x = \beta^+ u_y^+ - \beta^- u_y^-$$

at $\mathbf{x}_T$ to obtain

$$\mathbf{M} \begin{pmatrix} u_L^- \\ u_R^- \\ u_T^- \end{pmatrix} = \mathbf{N}\mathbf{u} + \mathbf{d}$$

for

$$\mathbf{M} = (m_{ij})_{i,j=1,2,3}$$

where

$$\begin{aligned} m_{11} &= \left(\beta^+ \frac{3 - 2\theta_L}{(1 - \theta_L)(2 - \theta_L)} + (\beta^- + [\beta]n_y^2) \frac{2\theta_L \theta_R + \theta_R^2}{\hat{\theta}} + [\beta]n_y n_x \frac{(\theta_R + \theta_R^2)\theta_L}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_L}, \\ m_{12} &= \left(\beta^- + [\beta]n_y^2 \frac{(\theta_L)^2}{\hat{\theta}} + [\beta]n_x n_y \frac{-(\theta_L)^3 + (\theta_L)^2}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_L}, \\ m_{13} &= \left([\beta]n_y n_x \frac{1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_L}, \\ m_{21} &= \left(\beta^- + [\beta]n_y^2 \frac{(\theta_R)^2}{\hat{\theta}} + [\beta]n_x n_y \frac{(\theta_R)^3 + (\theta_R)^2}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{22} &= \left(\beta^+ \frac{3 - 2\theta_R}{(1 - \theta_R)(2 - \theta_R)} + (\beta^- + [\beta]n_y^2) \frac{2\theta_L \theta_R + (\theta_L)^2}{\hat{\theta}} + [\beta]n_y n_x \frac{(\theta_L - (\theta_L)^2)\theta_R}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{23} &= \left(-[\beta]n_y n_x \frac{1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_R}, \\ m_{31} &= \left(-[\beta]n_x n_y \frac{(\theta_R)^2 + (\theta_R + (\theta_R)^2)\theta_T}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_T}, \\ m_{32} &= \left([\beta]n_x n_y \frac{(\theta_L)^2 + (-\theta_L + (\theta_L)^2)\theta_T}{\hat{\theta}} \right) \Big|_{\mathbf{x}=\mathbf{x}_T}, \\ m_{33} &= \left(\beta^+ \frac{3 - 2\theta_T}{(1 - \theta_T)(2 - \theta_T)} + (\beta^- + [\beta]n_x^2) \frac{2\theta_T + 1}{\theta_T(\theta_T + 1)} \right) \Big|_{\mathbf{x}=\mathbf{x}_T} \end{aligned}$$

and

$$\hat{\theta} = \theta_R(\theta_R + \theta_L)\theta_L.$$

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